

NEX OSS/BSS ISP Cloud Platform - Billing, OSS, NOC, CRM, SaaS Multi-Tenant

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Dokumen ini menggabungkan kebutuhan billing ISP, OSS/BSS, FreeRadius AAA, GIS Network, Work Order, Project Documentation, serta revisi Router Management sebagai single source of network topology. POP dan BTS tidak lagi dibuat sebagai modul terpisah; keduanya direpresentasikan melalui Router Role seperti POP Router dan BTS Router.

OSS/BSS

Billing ISP

SaaS Multi-Tenant

White Label

GIS Network

Work Order

FreeRadius AAA

Router Management

SNMP Monitoring

ROS Script Generator

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Bagian PRD utama

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Phase roadmap eksekusi

SaaS

Siap dijual ke ISP lain

Router-first

Router mewakili POP/BTS

01

1. Executive Summary

NEX OSS/BSS ISP Cloud Platform adalah platform terintegrasi untuk mengelola operasional ISP end-to-end: CRM, provisioning, billing, payment, monitoring, ticketing, inventory, GIS, work order, partner portal, customer portal, hingga multi-tenant SaaS untuk dijual kembali kepada ISP lain.

Kesalahan yang harus dihindari: membangun aplikasi ini sebagai invoice app biasa. Produk ini harus diperlakukan sebagai OSS/BSS platform. Kalau fondasi tenant, customer, service, billing, provisioning, audit, dan API tidak benar sejak awal, biaya rework akan besar ketika pelanggan dan tenant mulai bertambah.

2. Business Goals

Area	Target Bisnis
Finance	Invoice otomatis, payment reconciliation otomatis, AR aging jelas, revenue leakage terdeteksi.
NOC	Provisioning, suspend, unsuspend, monitoring, dan impact analysis lebih cepat.
Sales	Lead, survey, quotation, contract, dan handover ke teknisi terdokumentasi.
Management	Dashboard KPI realtime untuk revenue, customer, SLA, utilisasi jaringan, dan performa tim.
Product/SaaS	Platform dapat dijual ke ISP lain dengan white-label, tenant isolation, lisensi, dan usage metering.

3.1 In Scope

- CRM & Sales Pipeline
- Customer Management & KYC
- Contract Management & E-Signature
- Product & Service Management
- Billing, Invoice, Tax, Payment Gateway
- Suspend/Unsuspend Automation
- OSS: Router Management, Router Interface, Router Link, OLT, ONU, FreeRadius AAA/RADIUS, NAS Management, IPAM, VLAN, CID Generator
- Router Role: Core Router, Aggregation Router, Edge Router, PPPoE Router, BNG, Wireless Gateway, POP Router, BTS Router
- Monitoring: PRTG, LibreNMS, SNMP, GenieACS, Router SNMP Monitoring
- GIS Network Management: Router Layer, Customer Layer, Fiber Layer, ODC/ODP Layer
- Work Order Management
- Trouble Ticket & SLA Monitoring
- Inventory FO & Perangkat
- Partner Portal & Commission Engine
- Customer Portal & Mobile App
- WhatsApp Gateway 2 Arah
- Revenue Assurance
- Router Script Generator untuk MikroTik RouterOS v6 dan v7 agar router dapat terhubung ke FreeRadius
- Multi-Tenant SaaS, White Label, License Management, Marketplace, API Platform

3.2 Out of Scope Phase Awal

- Router Management sebagai modul terpisah dikeluarkan. POP direpresentasikan sebagai Router Role: POP Router.
- Router Management sebagai modul terpisah dikeluarkan. Router Role direpresentasikan sebagai Router Role: BTS Router atau Wireless Gateway.
- MikroTik API Integration dikeluarkan dari phase awal. Autentikasi pelanggan memakai FreeRadius sebagai source of truth.
- AI penuh untuk NOC dan billing - masuk phase lanjutan.
- Marketplace connector komersial lengkap - setelah core platform stabil.
- Multi-country tax engine - setelah pasar Indonesia matang.

Keputusan arsitektur v3.4: Router menjadi objek utama jaringan. POP dan Router Role tidak dibuat sebagai tabel/modul utama terpisah supaya data model lebih sederhana. Jika satu perangkat atau site berfungsi sebagai Router Role/cukup gunakan Router Role: BTS Router, POP Router, atau Wireless Gateway.

4. User Roles & Permission

Role	Hak Akses Utama
Platform Owner / NEX Super Admin	Kelola seluruh tenant, lisensi, audit global, revenue SaaS, marketplace, system configuration.
Tenant ISP Owner	Full access untuk tenant miliknya. Tidak dapat melihat tenant lain.
Tenant Admin	Operasional tenant: customer, layanan, billing, NOC, user, role, report.
Finance	Invoice, payment, tax, AR aging, credit note, report revenue.
NOC	Monitoring, provisioning, OLT/ONU, Radius, suspend/unsuspend, ticket teknis.
Sales	Lead, survey request, quotation, customer onboarding, pipeline.
Teknisi / Field Engineer	Work order, instalasi, maintenance, upload foto, BA digital, GPS check-in/out.
Partner / Reseller	Customer miliknya, komisi, ticket, invoice terkait, performa layanan.
Customer	Portal customer: invoice, payment, ticket, status layanan, kontrak, faktur.

5. Customer Lifecycle

1. Lead masuk dari Sales/Website/Partner/WhatsApp.
2. Sales membuat data prospect dan request survey.
3. Sistem membuat Survey Work Order.
4. Teknisi melakukan survey, upload foto, GPS, dan hasil feasible/not feasible.
5. Sales membuat quotation dan kontrak.
6. Customer menyetujui kontrak atau e-signature.
7. Sistem membuat Installation Work Order.
8. Teknisi melakukan instalasi dan upload BA digital.
9. NOC melakukan provisioning: CID, VLAN, IP, Radius/ONU/Router.
10. Billing membuat invoice/prorate sesuai tanggal aktivasi.
11. Customer aktif.
12. Jika overdue melewati policy, sistem suspend otomatis.
13. Jika payment diterima, sistem unsuspend otomatis.
14. Jika kontrak berakhir atau terminate, sistem melakukan proses terminasi dan asset recovery.

6. System Architecture

6.1 Domain Utama

- CRM Domain
- OSS Domain
- BSS/Billing Domain
- NOC & Monitoring Domain
- GIS Network Domain
- Work Order / Field Service Domain
- Partner & Reseller Domain
- Customer Experience Domain
- SaaS Multi-Tenant Platform Domain
- API & Integration Domain
- Security, Audit & Compliance Domain
- Analytics & AI Domain

6.2 Prinsip Arsitektur

- API-first agar web, mobile, partner portal, dan integrasi pihak ketiga memakai backend yang sama.
 - Multi-tenant sejak awal, bukan ditambah belakangan.
 - Tenant isolation wajib di level aplikasi, database schema, query, audit, dan storage.
 - Event-driven untuk billing, payment, suspend, notification, dan provisioning.
 - Audit log untuk semua aksi penting.
 - Integrasi perangkat jaringan harus asynchronous dan memakai job queue agar UI tidak lambat.
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7. CRM & Sales Module

- Lead Management: source, PIC sales, status, pipeline, notes, follow-up.
 - Survey Request: dibuat dari lead dan otomatis menghasilkan Survey Work Order.
 - Quotation: paket, harga, diskon, masa kontrak, SLA, instalasi, PPN.
 - Won/Lost Analysis: alasan menang/kalah, kompetitor, nilai deal.
 - Handover ke NOC/Teknisi: setelah deal closed dan dokumen lengkap.
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8. Customer Management & KYC

- Personal Customer: nama, NIK, email, HP, alamat, GPS, dokumen KTP.
 - Corporate Customer: nama perusahaan, NPWP, NIB, akta, PIC, alamat legal, alamat instalasi.
 - Customer Status: Prospect, Survey, Pending, Active, Suspend, Terminate.
 - KYC Workflow: upload dokumen, verifikasi admin, approval/reject, catatan revisi.
 - Document Repository: kontrak, PO, BA, KTP, NPWP, NIB, faktur pajak.
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9.1 Product Catalog

- Dedicated Internet
- Broadband Internet
- Metro Ethernet
- IP Transit
- Cloud Hosting
- Colocation
- Managed Service
- GPON/FTTx
- Wireless Broadband

9.2 Service Data

- CID/SID
- Customer
- Product
- Service Category
- Bandwidth
- SLA
- Router
- Router Interface
- Router Role
- OLT/ONU/Switch jika relevan
- VLAN
- IP Address
- NAS Device
- Radius Profile
- Radius Username
- Contract Reference
- Billing Status
- Provisioning Status
- Radius Status: Pending, Active, Suspended, Terminated, Sync Failed
- Topology Status: Mapped / Unmapped / Mapping Incomplete

9.3 Service Category Flags

Field	Type	Purpose
-------	------	---------

requires_router_mapping	BOOLEAN	Menentukan apakah service wajib dipetakan ke router.
requires_radius	BOOLEAN	Menentukan apakah service membutuhkan FreeRadius authentication.
requires_ip_assignment	BOOLEAN	Menentukan apakah service membutuhkan IP assignment.
requires_vlan	BOOLEAN	Menentukan apakah service membutuhkan VLAN.

Service Category	Router Mapping	Radius	IP	VLAN
Broadband Internet	Yes	Yes	Yes	Yes
Dedicated Internet	Yes	Optional	Yes	Yes
GPON FTTH	Yes	Yes	Yes	Yes
Wireless Internet	Yes	Yes	Yes	Optional
Metro Ethernet	Yes	No	Optional	Yes
IP Transit	Yes	No	Yes	Optional
Cloud Hosting	No	No	Yes	No
Colocation	No	No	Optional	No
Domain / SSL / Email Hosting	No	No	No	No

Router Mapping wajib hanya untuk layanan yang menggunakan akses jaringan pelanggan. Service non-network seperti domain, SSL, email hosting, cloud hosting, VPS, dan colocation tidak wajib memiliki router mapping.

10.1 Router Management

Router Management menjadi modul utama untuk merepresentasikan seluruh node jaringan. POP dan BTS tidak dibuat sebagai modul terpisah. Jika sebuah router/site berfungsi sebagai POP atau Router Role, sistem menggunakan field Router Role.

10.1.1 Business Objective

- Mengetahui pelanggan Internet terhubung ke router mana.
- Mengetahui interface pelanggan berada pada router mana.
- Mengetahui router yang menjadi NAS untuk FreeRadius.
- Menampilkan status router up/down.
- Menampilkan status SNMP monitoring router.
- Mengetahui dampak pelanggan jika router down.
- Mengukur utilisasi router untuk capacity planning.
- Menghasilkan script konfigurasi MikroTik RouterOS v6/v7 agar router dapat terhubung ke FreeRadius.

10.1.2 Router Role

- Core Router
- Aggregation Router
- Edge Router
- PPPoE Router
- BNG
- Wireless Gateway
- POP Router
- BTS Router

10.1.3 Router Data

Field	Description
Router ID	Unique router identifier.
Router Name	Nama router yang tampil di aplikasi.
Hostname	Hostname router.
Vendor	MikroTik, Juniper, Cisco, Huawei, Nokia, atau vendor lain.
Model	Device model.
Serial Number	Nomor seri perangkat.
Router Role	Core, Edge, Aggregation, PPPoE Router, BNG, Wireless Gateway, POP Router, BTS Router.

Site Name	Nama site/lokasi router.
Latitude / Longitude	GPS location.
Management IP	IP untuk monitoring/manajemen.
Loopback IP	Loopback address untuk routing/monitoring.
Radius NAS Identifier	NAS identifier yang dipakai FreeRadius.
Status	Online, Offline, Warning, Critical, Maintenance.
SNMP Status	Reachable, Unreachable, Auth Failed, Not Configured.

10.1.4 Router Interface Management

- Interface Name
- Interface Type: Physical / Logical / VLAN / Loopback
- Router Parent
- VLAN ID
- IP Address
- Capacity
- Status: Up / Down / Disabled
- Service Mapping
- Customer Count
- Traffic In / Traffic Out
- Error / Discard

10.1.5 Router Link Management

- Memetakan hubungan Router A ke Router B.
- Link type: Fiber, Metro Ethernet, MPLS, Wireless Backhaul.
- Data link: router_a_id, router_b_id, interface_a_id, interface_b_id, capacity, utilization, status.

10.1.6 Customer Mapping

```

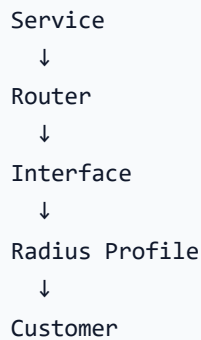
Customer
  ↓
Service
  ↓
Router
  ↓
Router Interface
  ↓
Radius NAS / FreeRadius

```

- Customer mapping menampilkan Customer, CID, Service, Router, Interface, Radius User, IP, dan VLAN.
- Jika service Internet belum memiliki router mapping, status service menjadi Mapping Incomplete.

- Service non-network tidak wajib memiliki router mapping.

10.1.7 Service Mapping



- Router mapping wajib hanya jika service category requires_router_mapping = true.
- Radius mapping wajib hanya jika service category requires_radius = true.
- Mapping dipakai oleh billing, Radius, NOC, GIS, monitoring, dan impact analysis.

10.1.8 Radius Integration



- NAS Device wajib dipetakan ke Router.
- Radius user wajib terhubung ke service dan customer jika requires_radius = true.
- Radius dashboard menampilkan Radius User per Router.
- Failed authentication dapat difilter berdasarkan Router/NAS.

10.1.9 SNMP Monitoring Integration

- Supported source: SNMP, LibreNMS, PRTG, Prometheus.
- Router status: Online, Offline, Warning, Critical, Maintenance.
- SNMP status: Reachable, Unreachable, Auth Failed, Not Configured.
- Metrics: CPU, memory, temperature, uptime, traffic, packet loss, latency.
- Interface metrics: traffic in, traffic out, error, discard, utilization.

10.1.10 Router Dashboard Requirement

Router Dashboard

Capacity Dashboard

- Total Router
- Online Router
- Offline Router
- Warning/Critical Router
- Top Utilization
- CPU Highest
- Memory Highest
- Customer per Router
- Radius User per Router

- Customer Count
- Service Count
- Active Session
- Interface Utilization
- CPU Utilization
- Memory Utilization
- Warning 70%
- Critical 85%
- Full 95%

10.1.11 Customer Impact Analysis

```

Router Down
↓
Find Service Mapping
↓
Find Customer
↓
Find Radius User
↓
Calculate Revenue Impact
↓
Show Active Ticket

```

- Router down menampilkan seluruh customer terdampak.
- Router down menampilkan affected service.
- Router down menampilkan affected radius user.
- Router down menampilkan total revenue terdampak.
- Router down menampilkan ticket aktif terkait customer terdampak.

10.1.12 MikroTik RouterOS RADIUS Script Generator

Sistem menyediakan generator script konfigurasi MikroTik yang dapat diunduh oleh user dalam format .rsc. Script ini digunakan agar router MikroTik dapat terhubung ke FreeRadius.

Input Script Generator

- Router Name
- RouterOS Version: v6 / v7
- Radius Server IP
- Radius Secret
- NAS Identifier
- Service Type: PPPoE / Hotspot
- Accounting Enabled: Yes / No

- Timeout

Output Script Generator

- Download ROS6 PPPoE Radius .rsc
- Download ROS7 PPPoE Radius .rsc
- Download ROS6 Hotspot Radius .rsc
- Download ROS7 Hotspot Radius .rsc
- Copy to Clipboard

Sample MikroTik ROS v6 PPPoE Radius Script

```
/radius
add service=ppp address={{radius_server_ip}} secret="{{radius_secret}}" authentication-
port=1812 accounting-port=1813 timeout=300ms

/ppp aaa
set use-radius=yes accounting=yes interim-update=5m

/system identity
set name="{{router_name}}"
```

Sample MikroTik ROS v7 PPPoE Radius Script

```
/radius
add service=ppp address={{radius_server_ip}} secret="{{radius_secret}}" authentication-
port=1812 accounting-port=1813 timeout=300ms

/ppp aaa
set use-radius=yes accounting=yes interim-update=5m

/system identity
set name="{{router_name}}"
```

Script harus dibuat dari template engine, bukan hardcode di source code. Gunakan entity router_script_templates agar nanti bisa menambah template RouterOS v6, RouterOS v7, JunOS, Huawei VRP, dan Cisco IOS tanpa perubahan besar pada aplikasi.

10.2 OLT & ONU Management

- Vendor support: ZTE, Huawei, Fiberhome, VSOL.
- OLT data: IP, vendor, router parent, SNMP/API credential, status.
- ONU data: serial number, ONU ID, PON, RX/TX power, status, customer, service.
- Fitur: discovery, provision, reboot, disable, monitoring optical power, alarm offline.

10.3 FreeRadius AAA & RADIUS Management

Modul ini mengelola Authentication, Authorization, dan Accounting pelanggan ISP menggunakan FreeRadius sebagai sumber autentikasi utama. MikroTik, BRAS, atau router NAS lain hanya berfungsi sebagai NAS/BNG yang melakukan query ke FreeRadius.

10.3.1 Supported Authentication

- PPPoE Username/Password Authentication
- Single Session dan Multi Session Control
- Simultaneous Use Control
- Session Timeout
- Idle Timeout
- Hotspot User Authentication
- Voucher Authentication
- MAC Authentication / MAC Binding
- Static IP Service Mapping
- VLAN Mapping untuk layanan tertentu

10.3.2 RADIUS Architecture

Layer	Fungsi
Customer Device	Router/ONT pelanggan melakukan koneksi PPPoE/Hotspot.
NAS / BNG	Router/NAS meneruskan authentication request ke FreeRadius.
FreeRadius	Melakukan authentication, authorization, dan accounting.
NEX OSS/BSS	Mengelola customer, service, billing status, radius user, radius profile, router mapping, dan suspend policy.
PostgreSQL	Database utama untuk customer, service, billing, radius profile, radius user, router mapping, dan accounting.

10.3.3 Radius Profile Management

- Profile Name
- Download Speed
- Upload Speed
- Burst Limit
- Burst Threshold
- Burst Time
- Session Timeout
- Idle Timeout
- Simultaneous Use
- Status: Active/Inactive

10.3.4 Radius User Management

- Auto generate username berdasarkan CID atau format tenant.
- Password dapat dibuat otomatis atau manual sesuai policy.

- Assign profile berdasarkan product/service.
- Status user: Pending, Active, Suspended, Terminated.
- Relasi wajib ke tenant, customer, service, dan router jika service membutuhkan akses jaringan.

10.3.5 NAS Device Management

- NAS Name
- NAS Identifier
- NAS IP Address
- Router Mapping
- Radius Secret
- Vendor Type
- Status

10.3.6 Service Activation Flow

1. Customer dan Service dibuat.
2. Sistem mengecek service category flags.
3. Jika requires_router_mapping = true, user wajib memilih Router dan Interface.
4. Jika requires_radius = true, Radius username dibuat otomatis.
5. Radius profile dipilih berdasarkan paket.
6. Radius user dibuat di database FreeRadius.
7. Service status berubah menjadi Active.

10.3.7 Suspend / Unsuspend Flow

- Invoice overdue melewati policy → Service Suspended → Radius User Disabled → pelanggan tidak dapat autentikasi.
- Invoice paid → Radius User Enabled → Service Active → pelanggan dapat autentikasi kembali.

10.4 IP Address Management (IPAM)

- IPv4/IPv6 pool, subnet, reservation, assignment customer, duplicate detection.
- Status IP: available, used, reserved, blocked.

10.5 VLAN Management

- VLAN registry, allocation, availability check, service mapping, router mapping.
- Mencegah VLAN bentrok antar layanan dan antar tenant.

10.6 CID Generator

Format rekomendasi: NEX-[AREA]-[SERVICE]-[YEAR]-[RUNNING]. Contoh: NEX-BDG-INET-2026-000001.

11.1 Billing Engine

- Recurring invoice bulanan.
- Prorate saat aktivasi tengah bulan.
- PPN, diskon, denda, deposit, credit note, debit note.
- Invoice status: Draft, Open, Paid, Overdue, Cancelled.

11.2 Payment Management

- Payment Gateway: Midtrans, Xendit, Duitku, QRIS, Virtual Account, transfer manual.
- Auto reconciliation dari callback payment gateway.
- Payment receipt dan payment history.

11.3 Suspend Engine

- Default suspend H+7 setelah jatuh tempo, configurable per tenant/customer/product.
- Auto unsuspend setelah payment verified.
- Whitelist customer strategis dengan approval finance.

11.4 Revenue Assurance

- Service aktif tanpa invoice.
- Invoice tanpa service aktif.
- VLAN/IP/ONU aktif tanpa billing.
- Customer aktif tanpa kontrak valid.
- Mismatch harga kontrak vs invoice.

11.5 Tax & e-Faktur

- PPN 11% configurable.
 - Export CSV e-Faktur/Coretax.
 - Faktur pajak per invoice.
 - Tax exemption untuk customer tertentu jika ada dokumen pendukung.
-

12. NOC & Monitoring Module

- NOC Dashboard 24x7: router online/offline, router warning/critical, customer down, ticket P1/P2/P3.
 - Monitoring integrations: PRTG API, LibreNMS API, SNMP, GenieACS.
 - RADIUS monitoring: FreeRadius availability, authentication success/failure, accounting status, online session summary.
 - Router monitoring: SNMP status, CPU, memory, interface traffic, utilization, latency, packet loss, uptime.
 - Customer Impact Analysis: saat router, OLT, ODP, fiber route, atau interface down, sistem menampilkan pelanggan terdampak dan revenue impact.
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13. Trouble Ticket & SLA Monitoring

- Kategori: Billing, Technical, Sales, General.
 - Status: Open, Assigned, In Progress, Resolved, Closed.
 - Priority: P1, P2, P3.
 - SLA: P1 4 jam, P2 8 jam, P3 24 jam - configurable per tenant.
 - Escalation: SLA countdown, breach alert, supervisor notification.
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14. GIS Network Management

GIS Network Management digunakan untuk memetakan infrastruktur jaringan ISP secara visual berbasis peta digital. Pada v3.4, Router menjadi node utama jaringan. POP dan BTS tidak menjadi layer wajib terpisah; keduanya direpresentasikan melalui Router Role.

14.1 Supported Map

- OpenStreetMap
- Google Maps
- Mapbox

14.2 GIS Objects

Object	Data Utama
Router	Router ID, hostname, role, GPS, site name, status, utilization.
ODC	ODC ID, lokasi, kapasitas core, available core.
ODP	ODP ID, lokasi, jumlah port, port terpakai, port tersedia.
Joint Closure	Closure ID, lokasi, jumlah core, core digunakan.
Fiber Route	Route ID, panjang kabel, jenis kabel, core count, status.
Customer	Customer ID, CID, service, lokasi GPS, router terkait.

14.3 GIS Features

- Layer: Router, Customer, Fiber, ODC, ODP.
- Router Role filter: Core Router, Aggregation Router, Edge Router, PPPoE Router, BNG, Wireless Gateway, POP Router, BTS Router.
- Fiber mapping: backbone, distribusi, dropcore.
- Core allocation, reservation, utilization, availability.
- Coverage area per router/site.
- Capacity planning ODP/ODC/core/router.
- Impact analysis: Router/OLT/ODP/backbone down - tampil pelanggan terdampak.

15. Work Order Management

Work Order Management mengelola pekerjaan lapangan dari survey sampai terminasi. Modul ini harus dibuat sebelum mobile app teknisi, karena mobile app hanya front-end dari workflow WO.

15.1 Work Order Categories

- Survey
- Instalasi
- Aktivasi
- Relokasi
- Upgrade
- Downgrade
- Maintenance
- Trouble Ticket
- Terminasi

15.2 Work Order Lifecycle

Open → Assigned → Accepted → On Progress → Completed → Verified → Closed

15.3 Field Service Mobile

- Melihat WO.
- Update status.
- Upload foto sebelum/saat/sesudah pekerjaan.
- Upload berita acara.
- GPS check-in/check-out.
- Customer signature dan technician signature.
- PDF generation untuk BA digital.

15.4 Automation

- Lead Won → Auto create Survey WO.
 - Survey Approved → Auto create Installation WO.
 - Installation Complete → Auto create Activation WO.
 - Technical Ticket → Auto create Trouble WO.
 - Termination Approved → Auto create Asset Recovery WO.
-

16. Inventory FO & Perangkat

- Asset category: router, switch, OLT, ONU, server, rack, SFP, radio, kabel FO, patch cord, closure.
- Serial number, asset number, vendor, purchase date, warranty, owner tenant, location, router assignment.
- Stock management, asset assignment, mutation history, asset recovery setelah terminasi.
- Router harus bisa menjadi inventory asset sekaligus network object.

17. Partner, Reseller & Commission

- Partner dapat melihat customer miliknya, invoice terkait, status layanan, ticket, dan komisi.
- Commission type: fixed, percentage, revenue sharing.
- Settlement report, paid/unpaid commission, commission history.
- Reseller SaaS: master reseller dan reseller dapat menjual tenant baru sesuai hak akses.

18. Customer Portal & Mobile App

- Customer melihat invoice, melakukan pembayaran, download invoice, download kontrak, download faktur pajak.
- Customer membuat ticket dan melihat status ticket.
- Customer melihat status layanan dasar: active/suspend/down/maintenance.
- Mobile app Android/iOS untuk invoice, payment, ticket, notification, service status.

19. WhatsApp Gateway 2 Arah

19.1 Outbound

- Invoice created
- Payment reminder
- Overdue notice
- Suspend notice
- Unsuspend notice
- Ticket update
- Maintenance notification

19.2 Inbound Command

- CEK TAGIHAN
- STATUS INTERNET
- OPEN TICKET
- CEK PEMBAYARAN
- BICARA CS

Inbound harus punya human handover. Jangan bikin chatbot sok pintar yang mengunci customer di loop otomatis. Itu bikin customer makin marah.

20. Contract Management & E-Signature

- Contract repository, start/end date, auto-renewal, amendment, versioning.
 - Reminder H-90, H-60, H-30, H-14, H-7.
 - E-signature dengan OTP verification, signature tracking, status Draft/Sent/Signed/Rejected.
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21. Multi-Tenant SaaS Architecture

Platform harus bisa digunakan banyak ISP dalam satu sistem dengan data terisolasi. Ini syarat utama jika produk akan dijual kembali.

21.1 Tenant Structure

Platform Owner (NEX) → Tenant ISP → Branch/Site → Customer

21.2 Tenant Data

- Tenant ID, company name, brand name, domain/subdomain, logo, NPWP, NIB, address, timezone, currency.
- Data tenant harus terisolasi: customer, billing, invoice, payment, router, OLT, ONU, inventory, partner, monitoring.

21.3 White Label

- Logo, favicon, app name, login page, dashboard theme.
- Custom domain: billing.isp-a.co.id, portal.isp-a.co.id, support.isp-a.co.id.
- SMTP, WhatsApp gateway, invoice template, contract template per tenant.

22. License Management & SaaS Billing

Package	Limit Awal
Trial	30 hari, maks 100 customer.
Starter	Maks 500 customer.
Professional	Maks 5.000 customer.
Enterprise	Unlimited sesuai kontrak.
Custom Enterprise	Dedicated deployment / on-premise / hybrid.

- Subscription: monthly, quarterly, annual, multi-year.
- Usage metering: customer count, router count, OLT count, ONU count, storage, API usage, WhatsApp usage, ticket count.
- Resource limitation mengikuti paket lisensi.

23. Marketplace & API Platform

23.1 Marketplace Addons

- WhatsApp Gateway
- SMS Gateway
- Payment Gateway
- FreeRadius Connector
- Router SNMP Monitoring Connector
- Huawei Connector
- ZTE Connector
- GenieACS Connector
- PRTG Connector
- LibreNMS Connector

MikroTik API tidak dimasukkan dalam marketplace phase awal. Fokus awal: FreeRadius Connector, Router SNMP Monitoring, dan Script Generator.

23.2 API Platform

- API key management: generate, revoke, expiry policy.
- OAuth2, JWT, OpenID Connect.
- Webhook events: customer created, invoice created, invoice paid, ticket created, service activated, service suspended, radius user created, radius user suspended, router down, router recovered.

24. Security, Audit & Compliance

- RBAC per tenant dan per role.
- 2FA untuk admin, finance, NOC, owner.
- Password policy dan session timeout.
- HTTPS only.
- Encryption at rest untuk data sensitif.
- Audit log: login, create, update, delete, suspend, unsuspend, payment, provisioning.
- Data privacy: tenant tidak boleh membaca data tenant lain. Ini non-negotiable.

25. Reporting & KPI

Kategori	Report/KPI
Financial	Revenue, MRR, ARR, AR Aging, cashflow, payment success rate.
Customer	Growth, churn rate, ARPU, active/suspend/terminate.
Network	Availability, utilization, packet loss, latency, router capacity.
NOC	MTTR, MTBF, SLA compliance, ticket breach, router outage impact.
Sales	Pipeline, conversion, win/loss, revenue per sales.
Partner	Commission, revenue share, customer growth.
SaaS	Tenant count, tenant retention, license revenue, add-on usage.

26. Recommended Tech Stack

Layer	Recommendation
Frontend Web	React / Next.js / Tailwind.
Mobile	Flutter untuk Android dan iOS.
Backend	Golang atau Laravel API. Untuk skala besar, Golang lebih kuat untuk service integrasi dan concurrency.
Database	PostgreSQL.
Cache	Redis.
Queue	RabbitMQ atau Kafka. Mulai dari RabbitMQ dulu agar tidak over-engineering.
Auth	Keycloak atau custom RBAC dengan OAuth2/OIDC.
Storage	S3-compatible storage untuk dokumen, foto, kontrak.
Monitoring	Prometheus, Grafana, Loki.
Deployment	Docker, Kubernetes untuk phase SaaS besar. Awal cukup Docker Compose/VM yang rapi.

- tenants
- users
- roles
- permissions
- customers
- customer_documents
- contracts
- service_categories
- products
- services
- service_resources
- invoices
- invoice_items
- payments
- tax_documents
- tickets
- work_orders
- devices
- routers
- router_interfaces
- router_links
- router_capacity_history
- router_script_templates
- olts
- onus
- ip_pools
- ip_addresses
- vlans
- fiber_routes
- odc
- odps
- joint_closures
- inventory_assets
- partners
- commissions
- notifications

- audit_logs
- api_keys
- webhooks
- license_plans
- tenant_subscriptions
- usage_metering

27.1 Database Entity Tambahan untuk FreeRadius AAA

Entity	Field Utama	Fungsi
radius_servers	id, tenant_id, name, ip_address, secret, status, created_at, updated_at	Master server FreeRadius per tenant.
nas_devices	id, tenant_id, radius_server_id, nas_name, nas_identifier, nas_ip_address, vendor_type, router_id, secret, status	Master NAS/BRAS/Router yang melakukan request ke FreeRadius.
radius_profiles	id, tenant_id, profile_name, download_speed, upload_speed, burst_limit, burst_threshold, burst_time, session_timeout, idle_timeout, simultaneous_use, status	Profile bandwidth dan policy akses pelanggan.
radius_users	id, tenant_id, customer_id, service_id, router_id, username, password_hash, profile_id, status, last_login, created_at, updated_at	User autentikasi PPPoE/Hotspot yang terhubung ke customer, service, dan router.
radius_accounting	id, tenant_id, radius_user_id, session_id, nas_identifier, framed_ip, login_time, logout_time, session_time, input_octets, output_octets, terminate_cause	Data accounting sesi pelanggan.
radius_sessions	id, tenant_id, radius_user_id, session_id, status, started_at, stopped_at, last_update_at	Monitoring session aktif dan histori session.
radius_sync_logs	id, tenant_id, entity_type, entity_id, action, status, error_message, created_at	Audit sinkronisasi data Radius.
radius_radcheck	id, username, attribute, op, value	Kompatibilitas struktur FreeRadius radcheck.
radius_radreply	id, username, attribute, op, value	Kompatibilitas struktur FreeRadius radreply.
radius_group_check	id, groupname, attribute, op, value	Policy check berbasis group/profile.
radius_group_reply	id, groupname, attribute, op, value	Reply attribute berbasis group/profile.

27.2 Database Entity Tambahan untuk Router Management

Entity	Field Utama	Fungsi
routers	id, tenant_id, router_name, hostname, vendor, model, serial_number, router_role, site_name, latitude, longitude, management_ip, loopback_ip, radius_nas_identifier, status, snmp_status, created_at, updated_at	Master router sebagai single source of network topology.

router_interfaces	id, tenant_id, router_id, interface_name, interface_type, vlan_id, ip_address, capacity, status, description	Master interface router dan titik service mapping.
router_links	id, tenant_id, router_a_id, router_b_id, interface_a_id, interface_b_id, link_type, capacity, status, utilization	Relasi antar router dan kapasitas link.
customer_router_mapping	id, tenant_id, customer_id, router_id, created_at	Mapping customer ke router.
service_router_mapping	id, tenant_id, service_id, router_id, interface_id, vlan_id, ip_address_id, radius_user_id	Mapping service ke router/interface.
router_capacity_history	id, tenant_id, router_id, cpu, memory, traffic, utilization, customer_count, service_count, session_count, captured_at	Historical capacity router.
router_script_templates	id, tenant_id, vendor, os_version, script_type, template_content, status, created_at, updated_at	Template generator script konfigurasi router, termasuk MikroTik ROS v6/v7.

27.3 Service Category Flags

Field	Type	Purpose
requires_router_mapping	BOOLEAN	Wajib router mapping atau tidak.
requires_radius	BOOLEAN	Wajib Radius user atau tidak.
requires_ip_assignment	BOOLEAN	Wajib IP assignment atau tidak.
requires_vlan	BOOLEAN	Wajib VLAN atau tidak.

28. Development Roadmap - Step by Step

Ini urutan yang harus diproses. Jangan mulai dari mobile app, GIS, atau AI. Itu terlihat keren, tapi kalau core data model dan billing engine belum benar, semua fitur keren hanya kosmetik mahal.

Phase	Prioritas	Output Utama
Phase 0 - Discovery & Blueprint	Paling awal	Finalisasi proses bisnis, ERD, arsitektur tenant, role matrix, API standard, UI wireframe.
Phase 1 - Core Foundation	Wajib	Auth, RBAC, tenant, customer, service category flags, product, service, CID generator, audit log.
Phase 2 - Billing MVP	Wajib	Invoice, recurring billing, prorate, payment manual, AR aging, customer portal basic.
Phase 3 - Payment & Notification	Tinggi	Payment gateway, QRIS/VA, WhatsApp/email notification, auto reconciliation.
Phase 4 - Router, FreeRadius AAA & Suspend	Tinggi	Router Management, Router Interface, Radius profile, Radius user, NAS management, RouterOS script generator, auto suspend/unsuspend.
Phase 5 - Ticket & Work Order	Tinggi	Ticketing, SLA, Survey WO, Installation WO, Activation WO, BA digital.
Phase 6 - Monitoring & NOC	Tinggi	Router SNMP monitoring, PRTG/LibreNMS/SNMP integration, FreeRadius monitoring, NOC dashboard, customer impact analysis.
Phase 7 - OSS Advanced	Sedang	OLT/ONU management, IPAM, VLAN management, router capacity planning.
Phase 8 - GIS & Inventory	Sedang	Mapping router/fiber/customer, asset tracking, FO core management.
Phase 9 - Partner & SaaS	Strategis	Partner portal, commission, multi-tenant license, white-label, usage metering.
Phase 10 - Marketplace & AI	Belakangan	Addon marketplace, AI NOC, anomaly detection, predictive churn.

MikroTik API Integration tidak masuk Phase 4. Yang masuk Phase 4 adalah RouterOS script generator untuk konfigurasi RADIUS, bukan kontrol router via API.

29. Detailed First 90 Days Plan

Week 1-2: Business Process & Data Blueprint

- Lock scope MVP: customer, service, billing, payment, suspend, notification, dan desain FreeRadius AAA.
- Buat flow actual Finance, NOC, Sales, Teknisi.
- Buat ERD awal dan role permission matrix.
- Tentukan tenant strategy: shared DB + tenant_id atau schema per tenant. Rekomendasi awal: shared DB dengan tenant_id dan row-level guard yang ketat.
- Tentukan flow Radius: service active, radius user active, invoice overdue, suspend, paid, unsuspend.

Week 3-4: UX/Wireframe & API Contract

- Wireframe dashboard, customer, service, invoice, payment, ticket, WO, radius profile, radius user, NAS device.
- Buat API contract OpenAPI/Swagger.
- Tentukan event: invoice.created, payment.paid, service.suspended, service.activated, radius.user.created, radius.user.suspended.

Week 5-8: Core Foundation Development

- Build auth, RBAC, tenant, customer, product, service, audit log.
- Build CID generator, status lifecycle, document upload.
- Build base dashboard internal.
- Build database awal untuk radius_profiles, radius_users, radius_servers, nas_devices.

Week 9-12: Billing MVP

- Build recurring invoice, prorate, invoice PDF, payment manual.
 - Build customer portal basic.
 - Build WhatsApp/email notification basic.
 - Build suspend flag di service dan radius user status, belum wajib disconnect session via API.
 - UAT dengan data customer NEX real sample.
-

30. Project Documentation Structure

Bagian ini menjadi panduan resmi struktur file dokumentasi proyek NEX OSS/BSS ISP Cloud Platform. Tujuannya agar PRD tidak berhenti sebagai dokumen fitur, tetapi menjadi blueprint eksekusi yang bisa dipakai product owner, developer, QA, AI coding assistant, project manager, dan tim operasional.

Brutal reality: kalau semua kebutuhan hanya ditumpuk dalam satu PRD besar, development akan cepat kacau. Setiap output kerja harus punya file sendiri, format sendiri, dan urutan pembuatan yang jelas.

30.1 Repository Structure

```
NEX-OSS-BSS/  
01-Business/  
├─ Customer-Lifecycle.md  
├─ Billing-Flow.md  
├─ Payment-Flow.md  
├─ Suspend-Flow.md  
├─ NOC-Flow.md  
├─ Ticket-Flow.md  
├─ WorkOrder-Flow.md  
├─ Partner-Commission-Flow.md  
├─ Contract-Flow.md  
├─ FreeRadius-Activation-Flow.md  
└─ Router-Router Role-Topology-Flow.md  
02-Architecture/  
├─ PRD-Master.html  
├─ PRD-Master.pdf  
├─ System-Architecture.md  
├─ Network-Topology.md  
├─ C4-Context.drawio  
├─ C4-Container.drawio  
├─ C4-Component.drawio  
└─ Event-Driven-Flow.drawio  
03-Database/  
├─ ERD.drawio  
├─ ERD.md  
├─ Database-Schema.sql  
├─ Migration-Plan.md  
├─ Role-Matrix.xlsx  
└─ Role-Matrix.md  
04-API/  
├─ openapi.yaml  
├─ swagger.json  
├─ postman-collection.json  
├─ API-Guideline.md  
├─ Auth-API.md  
└─ Customer-API.md
```

```

├─ Billing-API.md
├─ Radius-API.md
├─ Router-Router Role-API.md
├─ Ticket-API.md
05-UI-UX/
├─ Dashboard.fig
├─ Customer.fig
├─ Service.fig
├─ Billing.fig
├─ Radius.fig
├─ Router-Router Role.fig
├─ Ticket.fig
├─ NOC.fig

```

30.2 Document Creation Priority

Priority	Dokumen	Tujuan	Status Sebelum Coding
Priority 1	Business Flow	Menentukan perjalanan customer, billing, payment, suspend, ticket, WO, contract.	Wajib selesai.
Priority 1	Router-Router Role-Topology-Flow.md	Menentukan hubungan POP - Router - Router Role - Service - Customer.	Wajib selesai.
Priority 1	ERD.drawio / ERD.md	Fondasi database dan relasi antar domain.	Wajib selesai.
Priority 1	Role-Matrix.xlsx	Hak akses per role dan tenant isolation.	Wajib selesai.
Priority 1	openapi.yaml	Kontrak resmi backend, frontend, mobile, dan integrasi.	Wajib selesai.
Priority 2	Wireframe Figma	Acuan UI sebelum frontend dibuat.	Sebelum frontend development.

30.3 AI Development Workflow

```

PRD
↓
Business Flow
↓
Router Topology Flow
↓
ERD
↓
Role Matrix
↓
API Contract
↓
Wireframe
↓

```

OpenAPI Specification

↓

Generate Backend via AI / Codex

↓

Generate Frontend via AI / Codex

↓

Developer Review

↓

QA & UAT

↓

Production

30.4 Golden Rule Before Coding

Gate	Wajib Selesai	Alasan
Gate 1	Business Flow	Developer harus paham proses nyata, bukan hanya CRUD.
Gate 2	ERD	Database salah akan merusak billing, Radius, Router mapping, dan service lifecycle.
Gate 3	Role Matrix	Multi-tenant SaaS gagal jika permission dan tenant isolation tidak jelas.
Gate 4	API Contract	Frontend, backend, mobile, dan integrasi harus memakai kontrak yang sama.
Gate 5	Acceptance Criteria	Tanpa acceptance criteria, developer bisa merasa selesai padahal bisnis belum bisa jalan.

31. Acceptance Criteria

- User tenant tidak bisa melihat data tenant lain.
- Invoice bulanan otomatis dibuat sesuai billing cycle.
- Prorate berjalan untuk aktivasi tengah bulan.
- Payment mengubah status invoice menjadi Paid.
- Suspend otomatis berjalan sesuai policy.
- Unsuspend otomatis setelah payment verified.
- Service wajib memiliki Router Mapping hanya untuk service category yang menggunakan akses jaringan pelanggan.
- Service non-network seperti Cloud Hosting, VPS, Domain, SSL, Email Hosting, dan Colocation tidak wajib memiliki Router Mapping.
- Radius NAS wajib memiliki Router Mapping.
- Radius User wajib terhubung ke Customer, Service, Router, dan Radius Profile jika service membutuhkan authentication.
- Router dapat ditambahkan melalui menu Router.
- Router dapat menampilkan status Online, Offline, Warning, Critical, dan Maintenance.
- Router dapat dimonitor melalui SNMP.
- Router SNMP status dapat menampilkan Reachable, Unreachable, Auth Failed, dan Not Configured.
- Customer layanan Internet dapat ditelusuri sampai Router.
- Router Down menampilkan Customer Impact, Service Impact, Radius User Impact, Revenue Impact, dan Active Ticket.
- Router Dashboard tersedia.
- Capacity Dashboard tersedia.
- Radius Dashboard tersedia.
- Sistem dapat menghasilkan script MikroTik RouterOS v6 PPPoE Radius.
- Sistem dapat menghasilkan script MikroTik RouterOS v7 PPPoE Radius.
- Sistem dapat menghasilkan script MikroTik Hotspot Radius jika service membutuhkan Hotspot.
- Script dapat di-download dalam format .rsc dan dapat di-copy ke clipboard.
- CID, VLAN, IP, dan Radius Username tidak boleh duplikat dalam scope tenant yang sama.
- Ticket P1/P2/P3 memiliki SLA countdown dan escalation.
- WO bisa dibuat, di-assign, diselesaikan, diverifikasi, dan ditutup.
- GIS dapat menampilkan minimal Router, Customer, Fiber Route, ODP, dan ODC.
- Audit log mencatat semua aksi kritis.
- Multi-Tenant Isolation tetap berjalan untuk Router, Radius, Customer, Service, Billing, dan Monitoring.

32. Key Risks & Brutal Reality Check

- Risiko terbesar bukan coding. Risiko terbesar adalah data model buruk dan proses bisnis tidak dipaksa rapi.
- Jangan membangun semua modul sekaligus. Itu akan membuat proyek lambat, mahal, dan tidak selesai.
- Billing, customer, service, service category flags, router mapping, Radius status, tenant, role, dan audit adalah fondasi.
- POP dan Router Role sebagai modul terpisah akan memperberat model. Router Role lebih fleksibel dan cukup mewakili POP Router maupun BTS Router.
- FreeRadius harus menjadi source of truth untuk authentication phase awal. Jangan membuat logic dobel antara OSS/BSS, FreeRadius, dan router.
- MikroTik API Integration sengaja dikeluarkan dari phase awal agar MVP tidak melebar dan tidak terjebak masalah compatibility router.
- Script Generator lebih aman untuk fase awal dibanding API control, karena user tetap menginput konfigurasi ke router secara sadar.
- Integrasi OLT/ONU dan monitoring harus dibuat modular. Vendor command berbeda-beda; jangan hardcode ke satu vendor saja.
- SaaS multi-tenant harus dipikirkan dari hari pertama walaupun dijual ke ISP lain baru dilakukan belakangan.

33. Final Recommendation

Nama produk sebaiknya tidak lagi memakai kata Billing sebagai positioning utama. Nama yang lebih tepat: NEX ISP Cloud Platform, NEX OSS/BSS Cloud, NEXCore ISP Platform, atau NEX ISP Suite. Kata Billing terlalu sempit dan akan mengecilkan persepsi pasar terhadap nilai produk.

Urutan eksekusi paling rasional: Core Foundation → Billing MVP → Payment/Notification → Router Management + FreeRadius AAA → Ticket/WO → Router SNMP Monitoring/NOC → OSS Advanced → GIS/Inventory → SaaS/White Label → AI/Marketplace.

Untuk v3.4, Router menjadi single source of topology. POP dan BTS tidak dibuat sebagai modul terpisah. Gunakan Router Role: POP Router, BTS Router, Wireless Gateway, BNG, PPPoE Router, Edge Router, Aggregation Router, dan Core Router.

UPDATE PRD TERBARU

NEX OSS/BSS ISP Cloud Platform v3.4 - Router-Centric Update

Generated: 2026-06-19 | Source: PRD_NEX_OSS_BSS_Platform_v3.4_ROUTER_ONLY.html + Revisi Step 1-9

Bagian ini adalah konsolidasi terbaru dari gap analysis dan revisi Step 1 sampai Step 9. Prinsip arsitektur yang dikunci: **Customer -> Service -> Router -> Router Interface -> Radius NAS -> FreeRadius**. POP dan BTS tidak menjadi modul/tabel utama; keduanya hanya Router Role.

Router-Centric

FreeRadius AAA

SNMP Monitoring

Customer Impact Analysis

RouterOS Script Generator

UP

Kesimpulan Update Terbaru

- Router menjadi single source of network topology.
- Customer tidak terhubung langsung ke POP/BTS; customer ditelusuri melalui Service -> Router.
- Service Internet wajib memiliki Router Mapping; service non-network tidak wajib.
- Radius NAS dan Radius User wajib dapat ditelusuri ke Router untuk layanan jaringan.
- Router Down wajib menampilkan affected customer, affected service, affected radius user, revenue impact, dan incident.
- Roadmap mengikuti Phase 0 sampai Phase 10 dengan Phase 4 sebagai Router Management, FreeRadius AAA, dan Suspend Engine.
- UI wajib memiliki menu Network: Router, Router Interface, Router Link, Capacity Dashboard, Impact Analysis, dan SNMP Monitoring.

01 Step 1 - Business Flow**02** Step 2 - Database / ERD**03** Step 3 - Role Matrix**04** Step 4 - API Contract**05** Step 5 - Development Roadmap**06** Step 6 - Event Driven Architecture**07** Step 7 - State Machine**08** Step 8 - Wireframes**09** Step 9 - Acceptance Criteria

PRD NEX OSS/BSS - Flow Bisnis Utama v3.4 Router-Centric

Peran: Senior Business Analyst OSS/BSS ISP

Dokumen ini merevisi flow Step 1 agar konsisten dengan PRD v3.4. Prinsip topologi wajib:

Customer -> Service -> Router -> Router Interface -> Radius NAS -> FreeRadius

Router adalah node utama jaringan. POP dan BTS tidak menjadi modul/tabel utama; keduanya hanya nilai `router_role` seperti `POP Router` dan `BTS Router`.

Gap Analysis v3.4

Area	Gap Lama	Revisi v3.4
Topologi	Flow belum menempatkan router sebagai node utama	Semua provisioning dan NOC menelusuri dampak dari Service ke Router/Interface/NAS
POP/BTS	Berisiko dianggap modul utama	POP/BTS hanya role pada Router
FreeRadius	Aktivasi AAA belum wajib terkait router	Radius user wajib terkait service dan router untuk layanan jaringan
NOC	Incident belum menghitung dampak pelanggan dan revenue dari router down	Ditambahkan Customer Impact Analysis Flow
Provisioning	Belum ada flow khusus router	Ditambahkan Router Provisioning Flow

Aturan Umum Flow

- Billing hanya berjalan untuk customer dengan contract valid dan service aktif.
- Service Internet wajib memiliki Router Mapping.
- Service non-network tidak wajib Router Mapping.
- Radius user dibuat dari service yang membutuhkan AAA dan harus dapat ditelusuri ke router.
- Router Down wajib menghasilkan daftar customer terdampak, radius user terdampak, revenue impact, dan incident/ticket.

1. Customer Lifecycle

Diagram:

Prospect -> Lead -> Survey Work Order -> Feasible -> Quotation -> Contract Signed -> Installation Work Order -> Service Provisioning -> Router Mapping -> Radius User Active -> Billing Active -> Active Customer -> Suspend/Terminate

Status:

- prospect
- lead
- survey_requested

- `feasible` / `not_feasible`
- `quoted`
- `contract_signed`
- `installation_scheduled`
- `provisioning`
- `active`
- `suspended`
- `terminated`

Actor:

- Sales
- Customer
- Teknisi
- NOC
- Billing
- Partner jika customer berasal dari partner

Output:

- Customer master
- Contract
- Service instance
- Installation work order
- Router/service mapping
- Radius user jika layanan membutuhkan AAA
- Billing account

2. Billing Flow

Diagram:

`Service Active -> Validate Contract -> Validate Service Category -> Validate Router Mapping if Internet -> Generate Invoice -> Deliver Invoice -> Dunning -> Paid or Suspend`

Trigger:

- Billing cycle
- Service activation
- One-time installation charge
- Contract renewal or amendment

Validation:

- Service harus `active` .
- Service Internet wajib memiliki `service_router_mapping` .
- Invoice item harus mengacu ke service yang valid.
- Invoice overdue memicu Suspend Flow sesuai policy tenant.

Output:

- Invoice
- Invoice item
- AR aging
- `invoice.created`
- `invoice.overdue`
- `invoice.paid`

3. Payment Flow

Diagram:

`Invoice Issued -> Payment Initiated -> Payment Captured -> Reconciliation -> Invoice Paid -> Unsuspend Evaluation -> Service/Radius Reactivation`

Trigger:

- Payment gateway callback
- Manual payment
- Partner collection
- Customer portal payment

Output:

- Payment record
- Receipt
- Invoice status update
- Unsuspend event jika invoice terkait service suspended

4. Suspend / Unsuspend Flow

Diagram:

`Invoice Overdue -> Dunning Policy -> Suspend Service -> Disable Radius User -> Notify Customer -> Payment Reconciled -> Enable Radius User -> Unsuspend Service`

Rules:

- Suspend dilakukan pada service, bukan langsung pada customer global kecuali policy tenant mengatur full-account suspension.
- Service suspend wajib mencatat alasan, invoice, dan actor/system trigger.
- Radius user terkait service diubah menjadi `suspended`.
- Unsuspend hanya dilakukan setelah payment cleared atau approval manual.

Output:

- Service status `suspended` atau `active`
- Radius user status `suspended` atau `active`
- Audit log
- Notification

5. NOC Flow

Diagram:

Monitoring Alert -> Identify Router/Interface -> Customer Impact Analysis -> Incident Created -> Triage -> Escalation -> Work Order if Field Needed -> Resolution -> RCA -> Closure

Trigger:

- SNMP alert
- PRTG/LibreNMS alert
- Radius authentication anomaly
- Customer ticket
- Manual NOC report

Output:

- Incident
- Ticket
- Impact list
- Work order
- RCA

6. Ticket Flow

Diagram:

Ticket Created -> Categorize -> Link Customer/Service/Router -> Assign -> Investigate -> Resolve -> Customer Confirmation -> Close

Rules:

- Ticket teknis harus dapat mengacu ke customer, service, router, atau incident.
- Ticket dari Router Down dapat dibuat otomatis dari incident.
- Ticket billing tetap mengacu ke invoice/payment, tetapi dapat mengait ke service bila berdampak suspend.

7. Work Order Flow

Diagram:

Work Order Created -> Scope Router/Service/Customer -> Assign Technician -> Schedule -> Execute -> Upload Evidence -> Verify -> Close

Rules:

- Work order instalasi harus mengacu ke customer dan service.
- Work order network harus dapat mengacu ke router, router interface, atau router link.
- Completion report wajib memuat foto, GPS, checklist, dan hasil provisioning/repair.

8. Partner Commission Flow

Diagram:

Partner Sale -> Contract Signed -> Service Active -> Invoice Issued -> Payment Reconciled -> Commission Calculated -> Approval -> Payout

Rules:

- Komisi tidak dihitung dari invoice unpaid, cancelled, atau written-off.
- Komisi recurring mengacu ke payment yang sudah reconciled.
- Partner tidak mendapat akses ke router detail kecuali data ringkas layanan milik customer partner.

9. Contract Flow

Diagram:

Offer Accepted -> Draft Contract -> Review -> E-Signature -> Active Terms -> Service Order -> Renewal/Amendment/Termination

Rules:

- Contract signed wajib ada sebelum billing recurring aktif.
- SLA dan service category flags disimpan sebagai term yang memengaruhi provisioning.
- Contract dapat menentukan apakah service wajib router mapping dan radius user.

10. FreeRadius Activation Flow

Diagram:

Service Provisioning -> Validate Router Mapping -> Select Radius Profile -> Create Radius User -> Link Router/NAS -> Sync FreeRadius -> Test Authentication -> Activate Service

Rules:

- FreeRadius adalah source of truth untuk authentication phase awal.
- Radius user wajib mengacu ke customer, service, router, dan radius profile jika service membutuhkan akses jaringan.
- Radius NAS wajib mengacu ke router.
- Secret disimpan terenkripsi dan hanya ditampilkan saat creation/reset.

Output:

- Radius user
- Radius profile assignment
- NAS mapping
- Authentication test result
- radius.user.created

11. Router Provisioning Flow

Diagram:

Create Router -> Assign Router Role -> Create Interface -> Assign SNMP Monitoring -> Assign Radius NAS -> Map Service -> Map Customer

Router role:

- Core Router

- Aggregation Router
- Edge Router
- PPPoE Router
- BNG
- Wireless Gateway
- POP Router
- BTS Router

Rules:

- Router wajib memiliki tenant, hostname/router name, role, status, dan management IP.
- Interface dibuat di bawah router.
- SNMP dapat berstatus `Reachable` , `Unreachable` , `Auth Failed` , atau `Not Configured` .
- Radius NAS dibuat jika router berperan sebagai NAS/BNG/PPPoE/Hotspot gateway.

12. Customer Impact Analysis Flow

Diagram:

`Router Down -> Find Service Mapping -> Find Customer -> Find Radius User -> Calculate Revenue Impact -> Generate Incident`

Rules:

- Impact dihitung dari `service_router_mapping` , `customer_router_mapping` , dan radius user terkait.
- Revenue impact menggunakan MRR/recurring amount dari service aktif terdampak.
- Incident harus memuat severity, router, interface jika diketahui, customer count, service count, radius user count, dan revenue impact.

Output:

- Router impact report
- Incident
- Optional ticket/work order
- Customer notification batch

Dependency Antar Flow

- Customer Lifecycle -> Contract -> Work Order -> Router Provisioning -> FreeRadius Activation -> Billing.
- Billing -> Suspend -> Radius disable.
- Payment -> Unsuspend -> Radius enable.
- Router Provisioning -> NOC Monitoring.
- Router Down -> Customer Impact Analysis -> Incident -> Ticket/Work Order.
- Partner Commission bergantung pada Contract, Invoice, dan Payment Reconciled.

QC Checklist

- Tidak ada flow yang menghubungkan customer langsung ke POP atau BTS.
- Semua layanan jaringan melewati Service -> Router -> Router Interface.

- POP Router dan BTS Router hanya role pada Router.
 - Router Down menghasilkan customer impact, radius impact, revenue impact, dan incident.
-

PRD NEX OSS/BSS - Database Design v3.4 Router-Centric

Peran: Senior Database Architect

Tujuan dokumen ini adalah merevisi desain database Step 2 agar konsisten dengan PRD v3.4 Router-Centric.

Prinsip topologi wajib:

Customer -> Service -> Router -> Router Interface -> Radius NAS -> FreeRadius

POP dan BTS tidak boleh menjadi tabel/modul utama. Keduanya hanya direpresentasikan sebagai nilai `router_role` pada tabel `routers`.

Gap Analysis v3.4

Area	Gap Lama	Revisi v3.4
Router	Belum menjadi entity topology utama	Ditambahkan <code>routers</code> , <code>router_interfaces</code> , <code>router_links</code>
Mapping	Radius user hanya terkait service	Ditambahkan service/customer mapping ke router dan interface
Capacity	Belum ada histori kapasitas router	Ditambahkan <code>router_capacity_history</code>
Script	Belum ada template script RouterOS	Ditambahkan <code>router_script_templates</code>
POP/BTS	Berisiko menjadi tabel terpisah	Dilarang membuat <code>bts</code> , <code>bts_sites</code> , <code>pop</code> , <code>pop_sites</code>

Tabel Utama

- `tenants`
- `users`
- `roles`
- `permissions`
- `role_permissions`
- `user_roles`
- `customers`
- `products`
- `service_categories`
- `services`
- `contracts`
- `invoices`
- `invoice_items`
- `payments`
- `tickets`

- `work_orders`
- `radius_profiles`
- `radius_users`
- `nas_devices`
- `routers`
- `router_interfaces`
- `router_links`
- `customer_router_mapping`
- `service_router_mapping`
- `router_capacity_history`
- `router_script_templates`
- `audit_logs`

Tabel yang Dilarang

Tabel berikut tidak boleh dibuat karena digantikan oleh `routers.router_role` :

- `bts`
- `bts_sites`
- `pop`
- `pop_sites`

Relasi Wajib

- Customer -> Service
- Service -> Router melalui `service_router_mapping`
- Service -> Router Interface melalui `service_router_mapping.interface_id`
- Router -> Radius NAS melalui `nas_devices.router_id`
- Radius User -> Router melalui `radius_users.router_id`
- Radius User -> Service melalui `radius_users.service_id`
- Customer -> Router melalui `customer_router_mapping` untuk query impact cepat

Enum / Controlled Values

router_role

- `core_router`
- `aggregation_router`
- `edge_router`
- `pppoe_router`
- `bng`
- `wireless_gateway`
- `pop_router`

- `bts_router`

router_status

- `draft`
- `provisioning`
- `active`
- `warning`
- `critical`
- `maintenance`
- `offline`
- `retired`

snmp_status

- `reachable`
- `unreachable`
- `auth_failed`
- `not_configured`

router_interface_status

- `provisioning`
- `up`
- `down`
- `disabled`

radius_user_status

- `pending`
- `active`
- `suspended`
- `terminated`

SQL Skeleton

```
CREATE TABLE tenants (  
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),  
  name text NOT NULL,  
  org_number text,  
  plan text,  
  created_at timestamptz DEFAULT now(),  
  updated_at timestamptz  
);
```

```
CREATE TABLE customers (  
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
```

```

tenant_id uuid NOT NULL REFERENCES tenants(id),
external_ref text,
type text NOT NULL,
name text NOT NULL,
billing_contact jsonb,
status text NOT NULL DEFAULT 'prospect',
created_at timestamptz DEFAULT now(),
updated_at timestamptz,
UNIQUE (tenant_id, external_ref)
);

CREATE TABLE service_categories (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  code text NOT NULL,
  name text NOT NULL,
  requires_router_mapping boolean NOT NULL DEFAULT false,
  requires_radius boolean NOT NULL DEFAULT false,
  created_at timestamptz DEFAULT now(),
  UNIQUE (tenant_id, code)
);

CREATE TABLE products (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  service_category_id uuid REFERENCES service_categories(id),
  sku text NOT NULL,
  name text NOT NULL,
  pricing jsonb NOT NULL DEFAULT '{} '::jsonb,
  created_at timestamptz DEFAULT now(),
  UNIQUE (tenant_id, sku)
);

CREATE TABLE contracts (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  customer_id uuid NOT NULL REFERENCES customers(id),
  status text NOT NULL DEFAULT 'draft',
  start_date date,
  end_date date,
  terms jsonb NOT NULL DEFAULT '{} '::jsonb,
  signed_at timestamptz,
  created_at timestamptz DEFAULT now()
);

CREATE TABLE services (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  customer_id uuid NOT NULL REFERENCES customers(id),
  product_id uuid REFERENCES products(id),
  service_category_id uuid REFERENCES service_categories(id),
  contract_id uuid REFERENCES contracts(id),

```

```

cid text,
status text NOT NULL DEFAULT 'requested',
activated_at timestamptz,
suspended_at timestamptz,
terminated_at timestamptz,
metadata jsonb NOT NULL DEFAULT '{}'::jsonb,
created_at timestamptz DEFAULT now(),
updated_at timestamptz,
UNIQUE (tenant_id, cid)
);

CREATE TABLE routers (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  router_name text NOT NULL,
  hostname text NOT NULL,
  vendor text,
  model text,
  serial_number text,
  router_role text NOT NULL,
  site_name text,
  management_ip inet,
  latitude numeric(10,7),
  longitude numeric(10,7),
  status text NOT NULL DEFAULT 'draft',
  snmp_status text NOT NULL DEFAULT 'not_configured',
  snmp_profile jsonb,
  created_at timestamptz DEFAULT now(),
  updated_at timestamptz,
  deleted_at timestamptz,
  UNIQUE (tenant_id, hostname)
);

CREATE TABLE router_interfaces (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  router_id uuid NOT NULL REFERENCES routers(id),
  interface_name text NOT NULL,
  interface_type text,
  mac_address text,
  ip_address inet,
  vlan_id integer,
  speed_mbps integer,
  status text NOT NULL DEFAULT 'provisioning',
  created_at timestamptz DEFAULT now(),
  updated_at timestamptz,
  UNIQUE (tenant_id, router_id, interface_name)
);

CREATE TABLE router_links (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),

```

```

router_a_id uuid NOT NULL REFERENCES routers(id),
router_b_id uuid NOT NULL REFERENCES routers(id),
interface_a_id uuid REFERENCES router_interfaces(id),
interface_b_id uuid REFERENCES router_interfaces(id),
link_type text,
status text NOT NULL DEFAULT 'active',
capacity_mbps integer,
created_at timestamptz DEFAULT now()
);

CREATE TABLE service_router_mapping (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  service_id uuid NOT NULL REFERENCES services(id),
  router_id uuid NOT NULL REFERENCES routers(id),
  interface_id uuid REFERENCES router_interfaces(id),
  vlan_id integer,
  is_primary boolean NOT NULL DEFAULT true,
  created_at timestamptz DEFAULT now(),
  UNIQUE (tenant_id, service_id, router_id, interface_id)
);

CREATE TABLE customer_router_mapping (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  customer_id uuid NOT NULL REFERENCES customers(id),
  router_id uuid NOT NULL REFERENCES routers(id),
  created_at timestamptz DEFAULT now(),
  UNIQUE (tenant_id, customer_id, router_id)
);

CREATE TABLE radius_profiles (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  name text NOT NULL,
  attributes jsonb NOT NULL DEFAULT '{}'::jsonb,
  created_at timestamptz DEFAULT now(),
  UNIQUE (tenant_id, name)
);

CREATE TABLE nas_devices (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  router_id uuid NOT NULL REFERENCES routers(id),
  hostname text NOT NULL,
  nas_ip_address inet NOT NULL,
  vendor_type text,
  secret text NOT NULL,
  status text NOT NULL DEFAULT 'active',
  created_at timestamptz DEFAULT now(),
  UNIQUE (tenant_id, nas_ip_address)
);

```

```

CREATE TABLE radius_users (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  customer_id uuid NOT NULL REFERENCES customers(id),
  service_id uuid NOT NULL REFERENCES services(id),
  router_id uuid REFERENCES routers(id),
  profile_id uuid REFERENCES radius_profiles(id),
  username text NOT NULL,
  secret text NOT NULL,
  status text NOT NULL DEFAULT 'pending',
  created_at timestamptz DEFAULT now(),
  updated_at timestamptz,
  UNIQUE (tenant_id, username)
);

CREATE TABLE router_capacity_history (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  router_id uuid NOT NULL REFERENCES routers(id),
  cpu_percent numeric(5,2),
  memory_percent numeric(5,2),
  traffic_in_bps bigint,
  traffic_out_bps bigint,
  utilization_percent numeric(5,2),
  recorded_at timestamptz NOT NULL DEFAULT now()
);

CREATE TABLE router_script_templates (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid REFERENCES tenants(id),
  vendor text NOT NULL DEFAULT 'mikrotik',
  os_version text NOT NULL,
  script_type text NOT NULL,
  template_body text NOT NULL,
  variables_schema jsonb NOT NULL DEFAULT '{} '::jsonb,
  is_active boolean NOT NULL DEFAULT true,
  created_at timestamptz DEFAULT now(),
  UNIQUE (tenant_id, vendor, os_version, script_type)
);

```

ERD Teks

```

tenants
  +-- customers
  |   +-- services
  |       +-- service_router_mapping -- routers -- router_interfaces
  |       +-- radius_users -----+
  |
  +-- routers
  |   +-- router_interfaces

```

```
|    +-- router_links
|    +-- nas_devices
|    +-- router_capacity_history
|
+-- customer_router_mapping
+-- radius_profiles
+-- router_script_templates
```

Index dan Constraint Penting

- `services(tenant_id, customer_id, status)`
- `service_router_mapping(tenant_id, router_id)`
- `customer_router_mapping(tenant_id, router_id)`
- `radius_users(tenant_id, service_id)`
- `radius_users(tenant_id, router_id)`
- `router_capacity_history(tenant_id, router_id, recorded_at DESC)`
- `routers(tenant_id, router_role, status)`

Constraint aplikasi:

- Jika `service_categories.requires_router_mapping = true`, service aktif wajib memiliki `service_router_mapping`.
- Jika `service_categories.requires_radius = true`, service aktif wajib memiliki `radius_users`.
- Jika radius user dibuat untuk layanan jaringan, `radius_users.router_id` wajib terisi.
- `nas_devices.router_id` wajib mengacu ke router yang berperan sebagai NAS/BNG/PPPoE/Hotspot gateway.

Risiko dan Mitigasi

- Cross-tenant leakage: wajib RLS atau middleware tenant enforcement.
- Secret Radius/NAS bocor: simpan terenkripsi via KMS, jangan masuk log/plain backup.
- Impact query lambat: gunakan mapping table dan index router/customer/service.
- Data topology tidak sinkron: semua perubahan router/interface/mapping harus menghasilkan audit log dan event.

QC Checklist

- `routers`, `router_interfaces`, `router_links`, `customer_router_mapping`, `service_router_mapping`, `router_capacity_history`, dan `router_script_templates` tersedia.
 - Tidak ada tabel `bts`, `bts_sites`, `pop`, atau `pop_sites`.
 - Customer tidak terhubung langsung ke POP/BTS.
 - Service Internet wajib dapat ditelusuri ke router dan interface.
 - Radius NAS dan Radius User dapat ditelusuri ke router.
-

PRD NEX OSS/BSS - Role Permission Matrix v3.4 Router-Centric

Dokumen ini merevisi Step 3 agar seluruh permission mengikuti PRD v3.4 Router-Centric.

Legenda:

- Y = full allowed
- R = tenant-scoped/restricted
- S = self-only
- N = no access

Tenant-scoped roles hanya boleh membaca/mengubah data dengan `tenant_id` yang sama. Platform Owner dapat mengelola lintas tenant.

Gap Analysis v3.4

Area	Gap Lama	Revisi v3.4
Network permission	Belum ada Router Management lengkap	Ditambahkan Router, Interface, Link, SNMP, Capacity, Impact, Script Generator
Role NOC	Akses router belum eksplisit	NOC mendapat akses operasional router dan monitoring
POP/BTS	Berpotensi menjadi permission terpisah	Tidak ada permission POP/BTS; gunakan Router Role
Impact	Belum ada izin customer impact	Ditambahkan Customer Impact Analysis

Matrix

Module	Action/Permission	Platform Owner	Tenant Owner	Tenant Admin	Finance	NOC	Sales	Teknisi	Partner	Customer
Customer	Create customer	Y	R	R	N	N	R	N	R	N
Customer	Read customer	Y	R	R	R	R	R	R	R	S
Customer	Update customer	Y	R	R	N	N	R	N	R	S
Customer	Delete customer	Y	R	N	N	N	N	N	N	N
Service	Create service	Y	R	R	N	N	R	R	R	N
Service	Read service	Y	R	R	R	R	R	R	R	S
Service	Update service	Y	R	R	N	R	N	R	N	N

Module	Action/Permission	Platform Owner	Tenant Owner	Tenant Admin	Finance	NOC	Sales	Teknisi	Partner	Customer
Service	Suspend/unsuspend service	Y	R	R	N	R	N	N	N	N
Router Management	Create router	Y	R	R	N	R	N	N	N	N
Router Management	Read router	Y	R	R	N	R	N	R	N	N
Router Management	Update router	Y	R	R	N	R	N	N	N	N
Router Management	Retire/deactivate router	Y	R	R	N	R	N	N	N	N
Router Interface Management	Create interface	Y	R	R	N	R	N	N	N	N
Router Interface Management	Read interface	Y	R	R	N	R	N	R	N	N
Router Interface Management	Update interface	Y	R	R	N	R	N	N	N	N
Router Link Management	Create router link	Y	R	R	N	R	N	N	N	N
Router Link Management	Read router link	Y	R	R	N	R	N	R	N	N
Router Link Management	Update router link	Y	R	R	N	R	N	N	N	N
SNMP Monitoring	Configure SNMP	Y	R	R	N	R	N	N	N	N
SNMP Monitoring	Read SNMP status	Y	R	R	N	R	N	R	N	N
Capacity Dashboard	Read capacity	Y	R	R	R	R	N	R	N	N
Customer Impact Analysis	Run impact analysis	Y	R	R	R	R	N	R	N	N
Router Script Generator	Generate script	Y	R	R	N	R	N	N	N	N
Radius	Create profile	Y	R	R	N	R	N	N	N	N
Radius	Manage radius user	Y	R	R	N	R	N	R	N	N
Radius	Read accounting	Y	R	R	N	R	N	N	N	N

Module	Action/Permission	Platform Owner	Tenant Owner	Tenant Admin	Finance	NOC	Sales	Teknisi	Partner	Customer
Radius	Manage NAS device	Y	R	R	N	R	N	N	N	N
Billing	Generate invoice	Y	R	R	R	N	N	N	N	N
Billing	Read invoice	Y	R	R	R	N	N	N	R	S
Billing	Adjust invoice	Y	R	N	R	N	N	N	N	N
Payment	Record payment	Y	R	N	R	N	N	N	N	S
Payment	Reconcile payment	Y	R	N	R	N	N	N	N	N
Ticket	Create ticket	Y	R	R	N	R	R	R	R	S
Ticket	Read ticket	Y	R	R	N	R	R	R	R	S
Ticket	Update/resolve ticket	Y	R	R	N	R	N	R	N	N
Work Order	Create work order	Y	R	R	N	R	N	R	N	N
Work Order	Assign/schedule	Y	R	R	N	R	N	R	N	N
Work Order	Execute/report	Y	R	N	N	R	N	R	N	N
Contract	Draft contract	Y	R	R	N	N	R	N	R	N
Contract	Sign/activate contract	Y	R	R	N	N	R	N	R	N
Contract	Read contract	Y	R	R	R	N	R	N	R	S
Inventory	Read inventory	Y	R	R	N	R	N	R	R	N
Inventory	Update inventory	Y	R	R	N	R	N	R	N	N
GIS	View map	Y	R	R	N	R	N	R	R	N
GIS	Edit map	Y	R	R	N	R	N	R	N	N
Report	Run operational reports	Y	R	R	R	R	R	R	R	N
Audit Log	Read audit log	Y	R	R	N	N	N	N	N	N
Tenant Setting	Manage tenant settings	Y	R	R	N	N	N	N	N	N

Role yang Wajib Memiliki Akses Router-Centric

- Platform Owner: semua permission router, monitoring, impact, script.
- Tenant Owner: semua permission router dalam tenant.
- Tenant Admin: semua permission router operasional dalam tenant.
- NOC Technician/NOC: router management, interface, link, SNMP monitoring, capacity dashboard, customer impact analysis, router script generator.

Catatan Implementasi

- Tidak ada permission khusus POP/BTS. Gunakan permission Router Management dengan filter `router_role` .
 - Aksi generate script wajib masuk audit log karena dapat mengubah konfigurasi router saat diterapkan.
 - Finance boleh melihat capacity dan impact untuk estimasi revenue impact, tetapi tidak boleh mengubah router.
 - Partner dan Customer tidak boleh melihat detail topologi router tenant.
-

PRD NEX OSS/BSS - API Contract MVP v3.4 Router-Centric

Base path: `/api/v1`

Auth: Bearer token. Tenant-scoped request menggunakan tenant dari token atau path `{tenant_id}`. Semua endpoint harus menegakkan tenant isolation.

Gap Analysis v3.4

Area	Gap Lama	Revisi v3.4
Router endpoint	Belum tersedia	Ditambahkan <code>/routers</code> , <code>/router-interfaces</code> , <code>/router-links</code>
Impact/capacity/SNMP	Belum tersedia	Ditambahkan endpoint impact, capacity, SNMP
Script generator	Belum tersedia	Ditambahkan <code>POST /router-script-generator</code>
Radius mapping	Radius user belum eksplisit ke router	Request/response radius user memuat <code>router_id</code>

Common Errors

- 400: validation error
- 401: missing/invalid token
- 403: insufficient permission
- 404: resource not found
- 409: business conflict
- 500: server error

Auth

POST /auth/login

Request:

```
{"username":"string","password":"string"}
```

Response:

```
{"access_token":"string","token_type":"bearer","expires_in":3600,"refresh_token":"string"}
```

Customer

GET /tenants/{tenant_id}/customers

Permission: Tenant Owner, Tenant Admin, Sales, NOC restricted.

POST /tenants/{tenant_id}/customers

Permission: Tenant Owner, Tenant Admin, Sales, Partner restricted.

GET /tenants/{tenant_id}/customers/{customer_id}

Response includes service and router mapping summary for authorized internal roles.

Service

POST /tenants/{tenant_id}/services

Request:

```
{
  "customer_id": "uuid",
  "product_id": "uuid",
  "contract_id": "uuid",
  "service_category_id": "uuid",
  "cid": "CID-001",
  "metadata": {}
}
```

Validation:

- Customer, product, contract, and service category must belong to tenant.
- Signed/active contract is required for recurring billing.
- If service category requires router mapping, mapping must exist before activation.

PATCH /tenants/{tenant_id}/services/{service_id}

Allowed operations: activate, suspend, unsuspend, terminate, update metadata.

Router Management

GET /tenants/{tenant_id}/routers

Query params:

- `role`
- `status`
- `snmp_status`
- `q`
- `limit`
- `offset`

Permission: Platform Owner, Tenant Owner, Tenant Admin, NOC, Teknisi read-only.

Response model: `RouterStatus[]` .

POST /tenants/{tenant_id}/routers

Request:

```
{
  "router_name": "BNG Jakarta 01",
  "hostname": "bng-jkt-01",
  "vendor": "MikroTik",
  "model": "CCR",
  "serial_number": "string",
  "router_role": "bng",
  "site_name": "Jakarta POP",
  "management_ip": "10.0.0.1",
  "latitude": -6.200000,
  "longitude": 106.816666,
  "snmp_profile": {}
}
```

Validation:

- `router_role` must be one of Core Router, Aggregation Router, Edge Router, PPPoE Router, BNG, Wireless Gateway, POP Router, BTS Router.
- Hostname unique per tenant.

PUT /tenants/{tenant_id}/routers/{id}

Updates router identity, role, SNMP profile, status, or location.

GET /tenants/{tenant_id}/routers/{id}/impact

Response model: `RouterImpact` .

Returns customer count, service count, radius user count, revenue impact, and incident recommendation.

GET /tenants/{tenant_id}/routers/{id}/capacity

Response model: `RouterCapacity` .

Returns CPU, memory, traffic, utilization history, capacity warning, and capacity critical status.

GET /tenants/{tenant_id}/routers/{id}/snmp

Response model: `SNMPStatus` .

Returns `Reachable` , `Unreachable` , `Auth Failed` , or `Not Configured` .

Router Interface

GET /tenants/{tenant_id}/router-interfaces

Query params: `router_id` , `status` , `interface_type` .

POST /tenants/{tenant_id}/router-interfaces

Request:

```
{
  "router_id": "uuid",
  "interface_name": "ether1",
  "interface_type": "ethernet",
  "ip_address": "10.0.0.2",
  "vlan_id": 100,
  "speed_mbps": 1000
}
```

Router Link

GET /tenants/{tenant_id}/router-links

Query params: `router_id`, `status`, `link_type`.

POST /tenants/{tenant_id}/router-links

Request:

```
{
  "router_a_id": "uuid",
  "router_b_id": "uuid",
  "interface_a_id": "uuid",
  "interface_b_id": "uuid",
  "link_type": "backbone",
  "capacity_mbps": 10000
}
```

Service Router Mapping

POST /tenants/{tenant_id}/services/{service_id}/router-mapping

Request:

```
{
  "router_id": "uuid",
  "interface_id": "uuid",
  "vlan_id": 200,
  "is_primary": true
}
```

Validation:

- Required for Internet/network services.
- Optional for non-network services.
- Router and interface must belong to tenant.

Radius

POST /tenants/{tenant_id}/radius/profiles

Creates Radius profile.

POST /tenants/{tenant_id}/radius/users

Request:

```
{
  "customer_id": "uuid",
  "service_id": "uuid",
  "router_id": "uuid",
  "profile_id": "uuid",
  "username": "string",
  "secret": "string"
}
```

Validation:

- Username unique per tenant.
- Service must belong to customer and tenant.
- Router is required if service category requires router mapping.

POST /tenants/{tenant_id}/nas

Request:

```
{
  "router_id": "uuid",
  "hostname": "bng-jkt-01",
  "nas_ip_address": "10.0.0.1",
  "vendor_type": "mikrotik",
  "secret": "encrypted-or-plain-input"
}
```

Router Script Generator

POST /tenants/{tenant_id}/router-script-generator

Request:

```
{
  "router_id": "uuid",
  "os_version": "ROS7",
  "script_type": "PPPoE",
  "radius_server_ip": "10.10.10.10",
  "radius_secret": "string",
  "service_profile": "100M"
}
```

Response model: `GeneratedScript` .

Supported output:

- ROS6 PPPoE
- ROS7 PPPoE
- ROS6 Hotspot
- ROS7 Hotspot

Response Models

RouterStatus

```
{
  "id": "uuid",
  "router_name": "string",
  "hostname": "string",
  "router_role": "bng",
  "status": "active",
  "snmp_status": "Reachable",
  "management_ip": "10.0.0.1"
}
```

RouterCapacity

```
{
  "router_id": "uuid",
  "cpu_percent": 40.5,
  "memory_percent": 62.1,
  "traffic_in_bps": 1000000,
  "traffic_out_bps": 2000000,
  "utilization_percent": 70.2,
  "level": "normal|warning|critical"
}
```

RouterImpact

```
{
  "router_id": "uuid",
  "affected_customers": 120,
  "affected_services": 126,
  "affected_radius_users": 118,
  "revenue_impact_monthly": 25000000,
  "incident_severity": "P1"
}
```

SNMPStatus

```
{
  "router_id": "uuid",
```

```
"status": "Reachable",
"last_checked_at": "2026-06-18T00:00:00Z",
"latency_ms": 12
}
```

GeneratedScript

```
{
  "router_id": "uuid",
  "os_version": "ROS7",
  "script_type": "PPPoE",
  "script": "/radius add ...",
  "warnings": []
}
```

Billing, Payment, Ticket, Work Order

Existing MVP endpoints remain valid with these additions:

- Invoice generation validates service active state.
- Suspend triggers Radius user suspended.
- Ticket can reference `router_id`, `interface_id`, and `incident_id`.
- Work order can reference `router_id`, `interface_id`, and `router_link_id`.

QC Checklist

- Endpoint wajib ada: `GET /routers`, `POST /routers`, `PUT /routers/{id}`.
 - Endpoint wajib ada: `GET /router-interfaces`, `POST /router-interfaces`.
 - Endpoint wajib ada: `GET /router-links`, `POST /router-links`.
 - Endpoint wajib ada: `GET /routers/{id}/impact`, `GET /routers/{id}/capacity`, `GET /routers/{id}/snmp`.
 - Endpoint wajib ada: `POST /router-script-generator`.
 - Response model wajib ada: `RouterStatus`, `RouterCapacity`, `RouterImpact`, `SNMPStatus`, `GeneratedScript`.
-

PRD NEX OSS/BSS - Development Roadmap v3.4 Router-Centric

Peran: Technical Product Owner

Roadmap ini merevisi Step 5 agar mengikuti PRD v3.4. Urutan phase lama diganti penuh dengan Phase 0 sampai Phase 10 berikut.

Gap Analysis v3.4

Area	Gap Lama	Revisi v3.4
Phase 4	Fokus FreeRadius dan suspend saja	Menjadi Router Management, FreeRadius AAA, Suspend Engine
Monitoring	Belum eksplisit Router SNMP	Phase 6 fokus Monitoring & NOC dengan Router SNMP
GIS	Belum router layer sebagai dasar	Phase 8 GIS & Inventory memakai router/fiber/customer layer
Marketplace	Belum membatasi MikroTik API	Marketplace awal fokus FreeRadius Connector, Router SNMP, Script Generator

Phase Summary

Phase	Nama	Prioritas	Output Utama
Phase 0	Discovery & Blueprint	Wajib	Scope lock, process map, data model baseline, integration blueprint
Phase 1	Core Foundation	Wajib	Auth, RBAC, tenant, customer, product, service category flags
Phase 2	Billing MVP	Wajib	Invoice, recurring billing, prorate, AR aging, manual payment
Phase 3	Payment & Notification	Wajib	Payment gateway, reconciliation, WhatsApp/email notification
Phase 4	Router Management FreeRadius AAA Suspend Engine	Tinggi	Router, interface, NAS, Radius profile/user, RouterOS script, suspend/unsuspend
Phase 5	Ticket & Work Order	Tinggi	Ticket, SLA, work order, field report, installation/maintenance flow
Phase 6	Monitoring & NOC	Tinggi	Router SNMP, PRTG/LibreNMS, NOC dashboard, impact analysis
Phase 7	OSS Advanced	Sedang	OLT/ONU, IPAM, VLAN, capacity planning, revenue assurance
Phase 8	GIS & Inventory	Sedang	Router/customer/fiber map, ODC/ODP, asset tracking
Phase 9	Partner & SaaS	Strategis	Partner portal, commission, white label, license, usage metering
Phase 10	Marketplace & AI	Belakangan	Addon marketplace, AI NOC, anomaly detection, predictive churn

Phase 0 - Discovery & Blueprint

Objective:

- Mengunci scope MVP dan blueprint router-centric.

Deliverables:

- Business process map.
- Tenant and RBAC blueprint.
- Router-centric data model.
- Billing and payment policy.
- FreeRadius integration design.
- Monitoring integration decision.

Acceptance Criteria:

- POP/BTS disepakati sebagai Router Role, bukan modul utama.
- Service category flags `requires_router_mapping` dan `requires_radius` disetujui.
- Flow Radius: service active, radius user active, invoice overdue, suspend, paid, unsuspend disetujui.

Phase 1 - Core Foundation

Objective:

- Membangun fondasi SaaS multi-tenant.

Scope:

- Auth and session.
- Tenant management.
- RBAC and permission matrix.
- Customer master.
- Product catalog.
- Service category.
- Service instance.
- Audit log.

Acceptance Criteria:

- Data tenant terisolasi.
- Customer dan service dapat dibuat.
- Service category memiliki flags router/radius.
- Audit log mencatat aksi penting.

Phase 2 - Billing MVP

Objective:

- Menghasilkan billing MVP yang benar untuk ISP.

Scope:

- Invoice.
- Invoice items.
- Recurring billing.
- Prorate.
- AR aging.
- Manual payment record.
- Basic finance reports.

Acceptance Criteria:

- Invoice hanya dibuat untuk service valid.
- Invoice item mengacu ke service.
- Payment mengubah outstanding balance.
- AR aging dapat ditampilkan.

Phase 3 - Payment & Notification

Objective:

- Menghubungkan pembayaran dan komunikasi pelanggan.

Scope:

- Payment gateway integration.
- Payment webhook.
- Reconciliation.
- Receipt.
- WhatsApp/email notification.
- Dunning notification.

Acceptance Criteria:

- Payment reconciled dapat menandai invoice paid.
- Paid invoice dapat memicu unsuspend evaluation.
- Notifikasi invoice, overdue, paid, dan suspend terkirim.

Phase 4 - Router Management FreeRadius AAA Suspend Engine

Objective:

- Membangun inti OSS network activation berbasis router.

Scope:

- Router Management.
- Router Interface Management.
- Router Link Management.
- Radius NAS mapped to Router.
- Radius Profile.
- Radius User.

- Service Router Mapping.
- Customer Router Mapping.
- RouterOS Script Generator.
- Suspend/Unsuspend Engine.

Tasks Backend:

- CRUD `routers` , `router_interfaces` , `router_links` .
- CRUD `nas_devices` , `radius_profiles` , `radius_users` .
- Service activation validation for router mapping.
- Generate ROS6/ROS7 PPPoE and Hotspot scripts.
- Suspend service and disable Radius user.
- Unsuspend service and enable Radius user.

Tasks Frontend:

- Router Dashboard.
- Router Detail.
- Interface and link screens.
- Radius profile/user screens.
- Script Generator screen.
- Suspend/unsuspend controls.

Acceptance Criteria:

- Router dapat dibuat, diubah, dinonaktifkan.
- Router memiliki role Core, Aggregation, Edge, PPPoE, BNG, Wireless Gateway, POP Router, BTS Router.
- Service Internet wajib memiliki router mapping.
- Radius user aktif setelah service provisioning valid.
- Script generator menghasilkan ROS6 PPPoE, ROS7 PPPoE, ROS6 Hotspot, ROS7 Hotspot.
- Suspend mematikan Radius user; unsuspend mengaktifkannya kembali.

Phase 5 - Ticket & Work Order

Objective:

- Menghubungkan support, NOC, dan field operation.

Scope:

- Ticket.
- SLA.
- Work Order.
- Technician assignment.
- Installation report.
- Maintenance report.
- Photo/GPS evidence.

Acceptance Criteria:

- Ticket dapat mengacu ke customer, service, router, incident.
- Work order dapat mengacu ke router, interface, link, atau service.
- Field report dapat diverifikasi dan ditutup.

Phase 6 - Monitoring & NOC

Objective:

- Memberi visibility operasional jaringan dan dampak pelanggan.

Scope:

- Router SNMP Monitoring.
- Capacity Dashboard.
- PRTG/LibreNMS/SNMP integration.
- FreeRadius availability monitoring.
- NOC Dashboard.
- Customer Impact Analysis.
- Incident generation.

Acceptance Criteria:

- SNMP status menampilkan Reachable, Unreachable, Auth Failed, Not Configured.
- Router Down menampilkan customer terdampak, radius user terdampak, revenue impact.
- Capacity warning/critical memicu event.
- Incident dapat dibuat otomatis dari router down.

Phase 7 - OSS Advanced

Objective:

- Memperluas OSS setelah core router/radius stabil.

Scope:

- OLT/ONU management.
- IPAM.
- VLAN management.
- CID generator.
- Capacity planning.
- Revenue assurance.

Acceptance Criteria:

- VLAN dan CID tidak duplikat per tenant.
- IPAM dapat mengacu ke service/router/interface.
- Capacity planning membaca histori router.

Phase 8 - GIS & Inventory

Objective:

- Menyediakan peta dan inventory berbasis topology.

Scope:

- GIS Router Layer.
- GIS Customer Layer.
- Fiber Layer.
- ODC/ODP Layer.
- Inventory asset.
- Router as inventory asset and network object.

Acceptance Criteria:

- GIS menampilkan router, customer, fiber, ODC, ODP.
- Customer dapat ditelusuri ke router.
- Asset router dapat dikaitkan dengan object router.

Phase 9 - Partner & SaaS

Objective:

- Menyiapkan model partner dan SaaS multi-tenant untuk dijual ke ISP lain.

Scope:

- Partner portal.
- Commission engine.
- White label.
- License management.
- Usage metering.
- Tenant billing.

Acceptance Criteria:

- Komisi dihitung dari payment reconciled.
- Usage metering mencakup customer count, router count, storage, API usage, WhatsApp usage.
- Tenant isolation berlaku untuk router, radius, customer, service, billing, monitoring.

Phase 10 - Marketplace & AI

Objective:

- Menambahkan addon dan AI setelah platform stabil.

Scope:

- FreeRadius Connector.
- Router SNMP Monitoring Connector.
- Notification connector.
- AI NOC.
- Anomaly detection.

- Predictive churn.

Acceptance Criteria:

- Marketplace connector dapat diaktifkan per tenant.
- MikroTik API Integration tidak masuk phase awal.
- Script Generator tetap menjadi mekanisme RouterOS awal, bukan kontrol router via API.

Critical Path

1. Core Foundation harus selesai sebelum billing dan router.
2. Billing harus selesai sebelum suspend engine.
3. Router Management dan FreeRadius menjadi prasyarat NOC monitoring.
4. Ticket/Work Order menjadi prasyarat eskalasi incident operasional.
5. SaaS dan Marketplace dilakukan setelah tenant isolation stabil.

QC Checklist

- Roadmap memakai Phase 0 sampai Phase 10 sesuai PRD v3.4.
 - Phase 4 berisi Router Management, FreeRadius AAA, dan Suspend Engine.
 - Phase 6 berisi Router SNMP Monitoring, NOC, dan Impact Analysis.
 - POP/BTS tidak menjadi phase atau modul terpisah.
-

PRD NEX OSS/BSS - Event Driven Architecture v3.4 Router-Centric

Peran: Solution Architect OSS/BSS ISP

Dokumen ini merevisi Step 6 agar event system mendukung PRD v3.4 Router-Centric.

Gap Analysis v3.4

Area	Gap Lama	Revisi v3.4
Router events	Belum ada event router lifecycle	Ditambahkan <code>router.created</code> , <code>router.updated</code> , <code>router.deleted</code>
Monitoring	Belum ada router up/down/capacity/SNMP	Ditambahkan event monitoring router
Mapping	Service mapping belum event-driven	Ditambahkan <code>service.router.mapped</code>
Impact	Customer impact belum otomatis	Ditambahkan <code>customer.impact.detected</code> dan <code>incident.created</code>

Event Envelope

Semua event menggunakan envelope berikut:

```
{
  "event_id": "uuid",
  "event_name": "router.down",
  "schema_version": "1.0",
  "tenant_id": "uuid",
  "occurred_at": "2026-06-18T00:00:00Z",
  "trace_id": "uuid",
  "producer": "router-service",
  "payload": {}
}
```

Event Table

Event	Producer	Consumer	Payload Utama	Retry Strategy
<code>customer.created</code>	Customer Service	Billing, CRM, Audit	<code>customer_id</code> , <code>tenant_id</code> , <code>status</code>	3 retries, DLQ

Event	Producer	Consumer	Payload Utama	Retry Strategy
customer.updated	Customer Service	CRM, Audit, Impact Cache	customer_id , changed fields	3 retries, DLQ
service.created	Service Service	Provisioning, Billing, Audit	service_id , customer_id , category_id	3 retries, DLQ
service.activated	Service Service	Billing, Radius, Notification, Audit	service_id , customer_id , requires_radius	5 retries, DLQ
service.suspended	Service Service	Radius, Notification, Audit	service_id , reason , invoice_id	5 retries, DLQ
service.unsuspended	Service Service	Radius, Notification, Audit	service_id , reason , payment_id	5 retries, DLQ
service.router.mapped	Provisioning Service	Router, Radius, Billing, Audit	service_id , router_id , interface_id , vlan_id	5 retries, DLQ
router.created	Router Service	NOC, GIS, Audit	router_id , router_role , hostname , status	3 retries, DLQ
router.updated	Router Service	NOC, GIS, Impact Cache, Audit	router_id , changed fields	3 retries, DLQ
router.deleted	Router Service	NOC, GIS, Audit	router_id , deleted_at	3 retries, DLQ
router.up	Monitoring Service	NOC, Incident, Notification, Audit	router_id , snmp_status , recovered_at	5 retries, DLQ
router.down	Monitoring Service	Impact Analysis, NOC, Incident, Notification	router_id , detected_at , interface_id	10 retries, DLQ, alert on fail
router.capacity.warning	Capacity Service	NOC, Reporting, Notification	router_id , utilization_percent , metrics	5 retries, DLQ
router.capacity.critical	Capacity Service	NOC, Incident, Notification	router_id , utilization_percent , metrics	10 retries, DLQ, alert on fail
router.snmp.failed	Monitoring Service	NOC, Router Service, Audit	router_id , snmp_status , error	5 retries, DLQ
radius.user.created	Radius Service	FreeRadius Sync, Audit	radius_user_id , service_id , router_id	5 retries, DLQ
radius.user.suspended	Radius Service	FreeRadius Sync, Audit	radius_user_id , service_id , reason	5 retries, DLQ
radius.user.activated	Radius Service	FreeRadius Sync, Audit	radius_user_id , service_id	5 retries, DLQ
customer.impact.detected	Impact Analysis Service	Incident, NOC, Notification, Reporting	router_id , customer/service/radius counts, revenue impact	10 retries, DLQ

Event	Producer	Consumer	Payload Utama	Retry Strategy
incident.created	Incident Service	Ticket, Work Order, Notification, Audit	incident_id , severity , router_id , impact summary	5 retries, DLQ
invoice.created	Billing Service	Notification, Payment, Audit	invoice_id , customer_id , amount	3 retries, DLQ
invoice.paid	Payment Service	Billing, Service, Radius, Notification	invoice_id , payment_id , customer_id	5 retries, DLQ
payment.reconciled	Payment Service	Billing, Service, Radius, Audit	payment_id , invoice_id , amount	5 retries, DLQ
ticket.created	Ticket Service	NOC, Work Order, Notification, Audit	ticket_id , customer_id , router_id	3 retries, DLQ
work_order.created	Work Order Service	Technician, Notification, Audit	work_order_id , router_id , service_id	3 retries, DLQ

Router Down Cascade

```
router.down
-> Impact Analysis Service
-> customer.impact.detected
-> Incident Service
-> incident.created
-> Ticket Service / Work Order Service / Notification Service
```

Payload `customer.impact.detected` :

```
{
  "router_id": "uuid",
  "affected_customer_count": 120,
  "affected_service_count": 126,
  "affected_radius_user_count": 118,
  "revenue_impact_monthly": 25000000,
  "severity": "P1"
}
```

Service Activation Cascade

```
service.created
-> service.router.mapped
-> radius.user.created
-> service.activated
-> invoice schedule active
```

Rules:

- `service.router.mapped` wajib sebelum activation untuk layanan Internet.
- `radius.user.created` wajib untuk service dengan `requires_radius = true`.
- Billing recurring hanya aktif setelah service activation valid.

Capacity Warning Cascade

```
router.capacity.warning
-> NOC Dashboard
-> Capacity Dashboard
-> optional ticket
```

```
router.capacity.critical
-> Incident Service
-> NOC escalation
-> optional Work Order
```

Topic Naming

- `customer.events`
- `service.events`
- `router.events`
- `router.monitoring.events`
- `radius.events`
- `billing.events`
- `payment.events`
- `incident.events`
- `ticket.events`
- `work_order.events`

Retry and DLQ Rules

- Idempotency key: `event_id`.
- Consumers must be idempotent by target entity and event version.
- Retry uses exponential backoff.
- DLQ must preserve full envelope and error message.
- Monitoring events that fail after retry must alert NOC.

QC Checklist

- Event wajib ada: `router.created`, `router.updated`, `router.deleted`.
- Event wajib ada: `router.up`, `router.down`.
- Event wajib ada: `router.capacity.warning`, `router.capacity.critical`.
- Event wajib ada: `router.snmp.failed`.

- Event wajib ada: `service.router.mapped` .
 - Event wajib ada: `radius.user.created` .
 - Event wajib ada: `customer.impact.detected` .
 - Event wajib ada: `incident.created` .
 - Setiap event menampilkan producer, consumer, payload, dan retry strategy.
-

PRD NEX OSS/BSS - State Machine Diagrams v3.4 Router-Centric

Peran: Enterprise Solution Architect

Dokumen ini merevisi Step 7 agar state machine konsisten dengan PRD v3.4 Router-Centric.

Gap Analysis v3.4

Area	Gap Lama	Revisi v3.4
Router	Belum ada state machine Router	Ditambahkan Router: Draft sampai Retired
Router Interface	Belum ada state interface	Ditambahkan Provisioning, Up, Down, Disabled
Radius User	State lama tidak sama dengan PRD v3.4	Diganti menjadi Pending, Active, Suspended, Terminated
Service	Belum mewajibkan router mapping untuk Internet	Ditambahkan validation router mapping

Customer

States:

- prospect
- lead
- qualified
- contracted
- active
- suspended
- terminated

Transitions:

- prospect -> lead
- lead -> qualified
- qualified -> contracted
- contracted -> active
- active -> suspended
- suspended -> active
- active -> terminated
- suspended -> terminated

Validation:

- `contracted` requires signed contract.
- `active` requires at least one active service.
- `terminated` requires service cleanup and final billing handling.

Service

States:

- `requested`
- `provisioning`
- `active`
- `suspended`
- `terminated`
- `failed`

Transitions:

- `requested -> provisioning`
- `provisioning -> active`
- `provisioning -> failed`
- `active -> suspended`
- `suspended -> active`
- `active -> terminated`
- `suspended -> terminated`
- `failed -> terminated`

Validation:

- `active` requires contract and valid product/service category.
- Internet/network service requires `service_router_mapping`.
- Service with `requires_radius = true` requires active radius user.
- `suspended` requires reason and audit record.

Router

States:

- `Draft`
- `Provisioning`
- `Active`
- `Warning`
- `Critical`
- `Maintenance`
- `Offline`
- `Retired`

Transitions:

- Draft -> Provisioning
- Provisioning -> Active
- Provisioning -> Offline
- Active -> Warning
- Warning -> Active
- Warning -> Critical
- Critical -> Warning
- Critical -> Offline
- Active -> Maintenance
- Warning -> Maintenance
- Critical -> Maintenance
- Maintenance -> Active
- Active -> Offline
- Offline -> Active
- Offline -> Retired
- Maintenance -> Retired

Triggers:

- Draft -> Provisioning : router record completed.
- Provisioning -> Active : interface/SNMP/NAS setup completed.
- Active -> Warning : capacity warning or degraded SNMP.
- Warning -> Critical : capacity critical or severe packet loss.
- Critical -> Offline : router down.
- Active -> Offline : SNMP unreachable or manual outage.
- Offline -> Active : router recovered.
- Any non-retired -> Maintenance : planned maintenance.
- Offline/Maintenance -> Retired : decommission approved.

Validation:

- Provisioning requires hostname, router role, management IP, and tenant.
- Active requires at least one interface or explicit exception.
- Warning/Critical requires metric or monitoring evidence.
- Retired requires no primary active service mapping.

Router Interface

States:

- Provisioning
- Up
- Down

- Disabled

Transitions:

- Provisioning -> Up
- Provisioning -> Down
- Up -> Down
- Down -> Up
- Up -> Disabled
- Down -> Disabled
- Disabled -> Provisioning

Triggers:

- Interface created
- SNMP interface status update
- Admin disable/enable
- Repair completed

Validation:

- Up requires parent router not retired.
- Disabled requires admin reason.
- Interface used by active service cannot be disabled without override and impact confirmation.

Radius User

States:

- Pending
- Active
- Suspended
- Terminated

Transitions:

- Pending -> Active
- Active -> Suspended
- Suspended -> Active
- Active -> Terminated
- Suspended -> Terminated
- Pending -> Terminated

Triggers:

- Pending -> Active : FreeRadius sync and auth test successful.
- Active -> Suspended : service suspended or billing overdue.
- Suspended -> Active : payment reconciled or manual unsuspend.
- Active/Suspended -> Terminated : service termination.

Validation:

- **Active** requires customer, service, profile, and router when network service requires mapping.
- **Suspended** requires linked service suspend reason.
- **Terminated** requires service termination or cleanup approval.

Invoice

States:

- **draft**
- **issued**
- **overdue**
- **partial_paid**
- **paid**
- **written_off**
- **cancelled**

Transitions:

- **draft -> issued**
- **issued -> overdue**
- **issued -> partial_paid**
- **partial_paid -> paid**
- **issued -> paid**
- **overdue -> paid**
- **overdue -> written_off**
- **draft -> cancelled**
- **issued -> cancelled**

Validation:

- **issued** requires invoice number, due date, and items.
- **paid** requires paid amount equals total amount.
- **written_off** requires approval and audit.

Payment

States:

- **initiated**
- **authorized**
- **captured**
- **reconciled**
- **failed**
- **refunded**
- **cancelled**

Transitions:

- initiated -> authorized
- authorized -> captured
- captured -> reconciled
- initiated -> failed
- captured -> failed
- reconciled -> refunded
- authorized -> cancelled

Ticket

States:

- new
- acknowledged
- assigned
- in_progress
- resolved
- awaiting_customer
- closed
- escalated

Validation:

- Ticket from router incident must include `router_id` or `incident_id`.
- `resolved` requires resolution note.

Work Order

States:

- created
- planned
- scheduled
- executing
- completed
- verified
- closed
- cancelled

Validation:

- Network work order can reference router, interface, or router link.
- Completion requires report, photo/GPS evidence where field work applies.

Contract

States:

- draft
- negotiation
- signed
- active
- renewal_pending
- expired
- terminated

Validation:

- Service activation requires signed or active contract unless explicitly waived by tenant policy.

Summary

```
Router: Draft -> Provisioning -> Active -> Warning/Critical -> Offline -> Active -> Retired
Router Interface: Provisioning -> Up -> Down -> Up / Disabled
Radius User: Pending -> Active -> Suspended -> Active / Terminated
Service: requested -> provisioning -> active -> suspended -> active / terminated
```

QC Checklist

- Router state tersedia: Draft, Provisioning, Active, Warning, Critical, Maintenance, Offline, Retired.
 - Router Interface state tersedia: Provisioning, Up, Down, Disabled.
 - Radius User state tersedia: Pending, Active, Suspended, Terminated.
 - Service Internet tidak bisa aktif tanpa router mapping.
-

PRD NEX OSS/BSS - Wireframes v3.4 Router-Centric

Peran: Product Designer dan Business Analyst

Dokumen ini merevisi Step 8 agar UI mengikuti PRD v3.4 Router-Centric.

Gap Analysis v3.4

Area	Gap Lama	Revisi v3.4
Navigation	Belum ada menu Network router-centric	Ditambahkan Network dengan Router, Interface, Link, Capacity, Impact, SNMP
Router pages	Belum lengkap	Ditambahkan Router Dashboard, Detail, Capacity, Impact, Script Generator
Responsive	Belum mencakup halaman router	Ditambahkan desktop, tablet, mobile
POP/BTS	Berpotensi menjadi menu terpisah	POP/BTS hanya filter Router Role

Navigation

```
Dashboard
Customer
Service
Billing
Payment
Network
  Router
  Router Interface
  Router Link
  Capacity Dashboard
  Impact Analysis
  SNMP Monitoring
Radius
Ticket
Work Order
Inventory
GIS
Partner
Report
Settings
```

Tidak ada menu POP atau BTS terpisah. Gunakan filter `Router Role` pada halaman Router.

Global Desktop Shell

```

+-----+
| NEX OSS/BSS | Search... | Tenant | User |
+-----+
| Dashboard | Page Title | Actions |
| Customer | ----- |
| Service | Content Area |
| Billing | |
| Payment | |
| Network > | |
| Radius | |
| Ticket | |
| Work Order | |
| Settings | |
+-----+

```

Global Tablet Shell

```

+-----+
| Menu | NEX OSS/BSS | Search | User |
+-----+
| Page Title | Actions |
| ----- |
| Content in 2-column or stacked layout |
+-----+

```

Global Mobile Shell

```

+-----+
| Menu | NEX OSS/BSS | User |
+-----+
| Page Title |
| Primary Action |
| ----- |
| Stacked content |
+-----+

```

Router Dashboard

Purpose:

- Memberi ringkasan kesehatan router, capacity, SNMP, dan impact.

Desktop:

```

+-----+
| Router Dashboard | [+ Router] [Generate] |
+-----+
| Total Router | Active | Warning | Critical | Offline | SNMP Failed |

```

Filters: Role [All] Status [All] SNMP [All] Search [_____]						
Router Table						
Hostname	Role	Status	SNMP	Utilization	Impact	Actions
bng-jkt-01	BNG	Active	Reachable	64%	0	View
pop-bdg-01	POP Router	Warning	Reachable	82%	45	View
bts-01	BTS Router	Offline	Unreachable	-	120	View

Tablet:

Router Dashboard				[+]
Total	Active	Warning	Offline	
Filters				
Router list cards				
bng-jkt-01	BNG	Active	64%	View

Mobile:

Router Dashboard		+
Search		
Role	Status	
bng-jkt-01		
BNG	Active	Reachable
Utilization	64%	Impact 0

Router Detail

Purpose:

- Menampilkan identitas router, role, status, interface, NAS, service mapping, dan event timeline.

Desktop:

Router Detail: bng-jkt-01						[Edit] [Maintenance] [Retire]
Summary: Role BNG Status Active SNMP Reachable IP 10.0.0.1 Site Jakarta						
Interfaces			Radius NAS			
ether1	Up	1G VLAN 100	NAS IP	10.0.0.1	Vendor MikroTik	Status Active
ether2	Up	1G VLAN 200	Secret:	*****		
Service Mapping						
CID	Customer	Service	Interface	VLAN	Status	

Capacity Dashboard

Role [All] Site [All]

Top Utilization

CPU Warning

Memory Warning

Traffic Critical

Router Utilization Chart

Router	CPU	Memory	In/Out Traffic	Utilization	Level
pop-bdg-01	72%	68%	900M/850M	82%	Warning

Mobile:

Capacity			
pop-bdg-01			
CPU 72%	Mem 68%	82% Warn	

Router Impact

Purpose:

- Menghitung customer/service/radius/revenue impact dari router down.

Desktop:

+-----+

| Impact Analysis

Router [bts-01] [Run Analysis] |

+-----+

| Affected Customers | Affected Services | Radius Users | Monthly Revenue Impact |

| 120 | 126 | 118 | IDR 25,000,000 |

+-----+

| Impacted Customers

| Customer CID Service Radius User Status MRR |

+-----+

| [Create Incident] [Create Ticket] [Export]

+-----+

Mobile:

Impact Analysis	
Router: bts-01	
Customers 120	
Services 126	
Radius Users 118	
Revenue IDR 25,000,000	

```
| Create Incident |
+-----+
```

Router Script Generator

Purpose:

- Menghasilkan script RouterOS untuk integrasi FreeRadius tanpa Mikrotik API.

Desktop:

```
+-----+
| Router Script Generator |
+-----+
| Router [bng-jkt-01] OS [ROS7] Type [PPPoE] Radius Server [10.10.10.10] |
| Secret [*****] Service Profile [100M] [Generate] |
+-----+
| Generated Script |
| /radius add address=10.10.10.10 ... |
| |
| [Copy] [Download] [Audit Log] |
+-----+
```

Supported script:

- ROS6 PPPoE
- ROS7 PPPoE
- ROS6 Hotspot
- ROS7 Hotspot

Mobile:

```
+-----+
| Script Generator |
| Router |
| OS Version |
| Script Type |
| Radius Server |
| Secret |
| Generate |
| Script output |
+-----+
```

SNMP Monitoring Dashboard

```
+-----+
| SNMP Monitoring | Status [All] [Refresh] |
+-----+
| Reachable | Unreachable | Auth Failed | Not Configured |
+-----+
```

Router	Status	Last Check	Latency	Action
bng-jkt-01	Reachable	2026-06-18 09:00	12 ms	Detail
pop-bdg-01	Auth Failed	2026-06-18 09:00	-	Fix SNMP

Radius Screen Update

Radius
Profiles Users NAS Devices
Radius Users: Username Customer Service Router Profile Status
NAS Devices: Hostname Router NAS IP Vendor Status

NOC Dashboard Update

NOC Dashboard
Router Online Router Warning Router Critical Customer Down Open P1
Active Incidents
Incident Router Impact Customer Revenue Impact SLA Action

GIS Update

Layers:

- Router
- Customer
- Fiber
- ODC
- ODP

Filters:

- Router Role
- Status
- SNMP Status
- Customer Impact

Responsive Rules

- Desktop uses sidebar navigation and wide tables.

- Tablet uses collapsible navigation and two-column panels.
- Mobile uses stacked cards, sticky primary action, and tabs for dense detail.
- Router tables collapse into summary cards on mobile.
- Impact Analysis keeps metrics visible before detailed list on all viewports.

QC Checklist

- Menu Network tersedia dengan Router, Router Interface, Router Link, Capacity Dashboard, Impact Analysis, SNMP Monitoring.
 - Halaman tersedia: Router Dashboard, Router Detail, Router Capacity, Router Impact, Router Script Generator, SNMP Monitoring Dashboard.
 - Desktop, tablet, dan mobile tersedia.
 - Tidak ada menu POP/BTS terpisah.
-

PRD NEX OSS/BSS - MVP Acceptance Criteria v3.4 Router-Centric

Peran: QA Lead dan Technical Product Owner

Dokumen ini merevisi Step 9 agar acceptance criteria konsisten dengan PRD v3.4.

Gap Analysis v3.4

Area	Gap Lama	Revisi v3.4
Router	Belum menjadi acceptance criteria utama	Ditambahkan CRUD router, role, SNMP, impact, capacity
Service	Router mapping belum wajib untuk Internet	Ditambahkan rule wajib mapping untuk service Internet
Radius	Radius belum dikaitkan ke router	Radius user wajib terkait service dan router jika layanan jaringan
Script	Belum ada output script generator	Ditambahkan ROS6/ROS7 PPPoE/Hotspot

Customer and Service

Feature: Customer and Service Lifecycle

Acceptance Criteria:

- Customer dapat dibuat, diubah, dibaca, dan dinonaktifkan sesuai role.
- Service dapat dibuat untuk customer dengan product dan contract valid.
- Service Internet wajib memiliki Router Mapping sebelum status `active`.
- Service non-network tidak wajib Router Mapping.
- Service dengan `requires_radius = true` wajib memiliki Radius User.
- Customer tidak pernah dihubungkan langsung ke POP/BTS; relasi selalu melalui Service -> Router.

Test Cases:

Scenario	Expected Result
Activate Internet service tanpa router mapping	Ditolak dengan validation error
Activate non-network service tanpa router mapping	Berhasil jika rule lain valid
Read customer impact summary	Menampilkan service dan router terkait untuk role internal

Router Management

Feature: Router CRUD and Role

Acceptance Criteria:

- Router dapat dibuat, diubah, dan dinonaktifkan.
- Router memiliki role:
 - Core Router
 - Aggregation Router
 - Edge Router
 - PPPoE Router
 - BNG
 - Wireless Gateway
 - POP Router
 - BTS Router
- Hostname router unik per tenant.
- Router dapat memiliki interface.
- Router dapat memiliki router link.
- Router dapat dikaitkan ke Radius NAS.
- POP dan BTS tidak tersedia sebagai tabel/modul terpisah.

Test Cases:

Scenario	Expected Result
Create router role POP Router	Router tersimpan sebagai <code>router_role</code> , bukan POP entity
Create duplicate hostname dalam tenant sama	Ditolak dengan conflict
Deactivate router yang masih punya primary active service mapping	Ditolak atau butuh override approval

Router Interface and Link

Acceptance Criteria:

- Router Interface dapat dibuat, diubah, dibaca.
- Router Interface memiliki state Provisioning, Up, Down, Disabled.
- Router Link dapat menghubungkan dua router dan optional interface.
- Interface yang digunakan active service tidak boleh disabled tanpa impact confirmation.

SNMP Monitoring

Acceptance Criteria:

- SNMP Status dapat ditampilkan:
 - Reachable
 - Unreachable
 - Auth Failed
 - Not Configured
- Router SNMP status dapat diperbarui oleh monitoring service.
- SNMP failed menghasilkan event `router.snmp.failed` .
- Router Down menghasilkan event `router.down` .

Test Cases:

Scenario	Expected Result
SNMP credential salah	Status <code>Auth Failed</code>
Router tidak bisa dihubungi	Status <code>Unreachable</code> , router menjadi Offline/Critical sesuai rule
SNMP belum dikonfigurasi	Status <code>Not Configured</code>

Capacity Dashboard

Acceptance Criteria:

- Dashboard menampilkan CPU, memory, traffic, utilization.
- Utilization warning menghasilkan `router.capacity.warning` .
- Utilization critical menghasilkan `router.capacity.critical` .
- Data histori tersimpan pada `router_capacity_history` .

Customer Impact Analysis

Acceptance Criteria:

- Router Down wajib menampilkan customer terdampak.
- Router Down wajib menampilkan service terdampak.
- Router Down wajib menampilkan radius user terdampak.
- Router Down wajib menampilkan revenue impact.
- Router Down wajib dapat menghasilkan incident.

Test Cases:

Scenario	Expected Result
Router down dengan 100 service aktif	Impact menampilkan 100 service dan customer terkait
Router down dengan radius users	Impact menampilkan radius user terdampak
Impact calculated	Revenue impact bulanan dihitung dari service aktif terdampak

Radius and FreeRadius AAA

Acceptance Criteria:

- Radius profile dapat dibuat dan diubah.
- Radius user dapat dibuat untuk service.
- Radius user memiliki state Pending, Active, Suspended, Terminated.
- Radius User wajib terhubung ke Customer, Service, Router, dan Radius Profile jika service membutuhkan akses jaringan.
- Radius NAS wajib memiliki Router Mapping.
- Suspend service mengubah Radius User menjadi Suspended.
- Unsuspend service mengubah Radius User menjadi Active.
- Secret disimpan terenkripsi dan tidak tampil ulang kecuali saat creation/reset.

Router Script Generator

Acceptance Criteria:

- Script Generator wajib menghasilkan:
- ROS6 PPPoE
- ROS7 PPPoE
- ROS6 Hotspot
- ROS7 Hotspot
- Output script memuat konfigurasi Radius server, secret, dan service profile sesuai input.
- Generate script wajib masuk audit log.
- MikroTik API Integration tidak wajib pada MVP phase awal.

Billing, Payment, Suspend

Acceptance Criteria:

- Invoice dapat dibuat untuk service aktif.
- Payment reconciled mengubah invoice menjadi paid.
- Invoice overdue dapat memicu suspend.
- Payment paid dapat memicu unsuspend.
- Suspend/unsuspend harus sinkron dengan Radius user.

Ticket and Work Order

Acceptance Criteria:

- Ticket dapat dibuat dari customer, NOC alert, atau incident.
- Ticket teknis dapat mengacu ke router, interface, service, dan customer.
- Work Order dapat dibuat dari ticket atau incident.
- Work Order field report memuat bukti pekerjaan.

Event-Driven Acceptance

Acceptance Criteria:

- Event router lifecycle tersedia: `router.created` , `router.updated` , `router.deleted` .
- Event monitoring tersedia: `router.up` , `router.down` , `router.capacity.warning` , `router.capacity.critical` , `router.snmp.failed` .
- Event mapping tersedia: `service.router.mapped` .
- Event impact tersedia: `customer.impact.detected` , `incident.created` .
- Semua consumer idempotent dan memiliki retry/DLQ.

Security and Tenant Isolation

Acceptance Criteria:

- Semua entity tenant-scoped memiliki `tenant_id` .
- User tenant tidak dapat membaca router/radius/customer tenant lain.

- Platform Owner dapat mengakses lintas tenant untuk operasi platform.
- Semua aksi sensitif masuk audit log.

Final MVP Gate

MVP v3.4 dianggap lolos jika:

- Service Internet bisa diaktifkan end-to-end sampai Router Mapping dan Radius User.
 - Router Down dapat dihitung dampaknya sampai customer, radius user, revenue impact, dan incident.
 - Billing, payment, suspend, dan unsuspend sinkron dengan Radius.
 - Tidak ada konsep POP/BTS sebagai modul utama atau tabel terpisah.
-